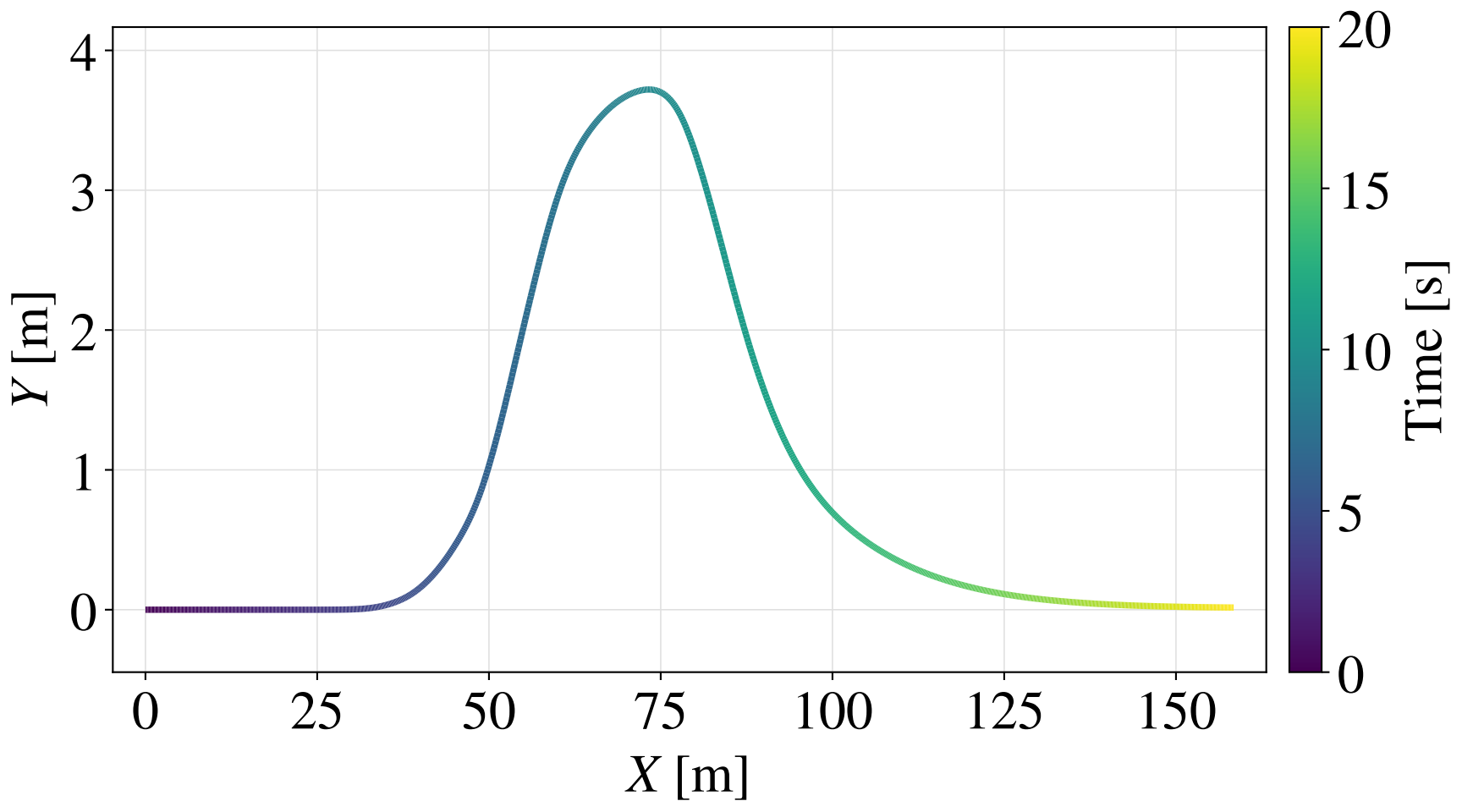
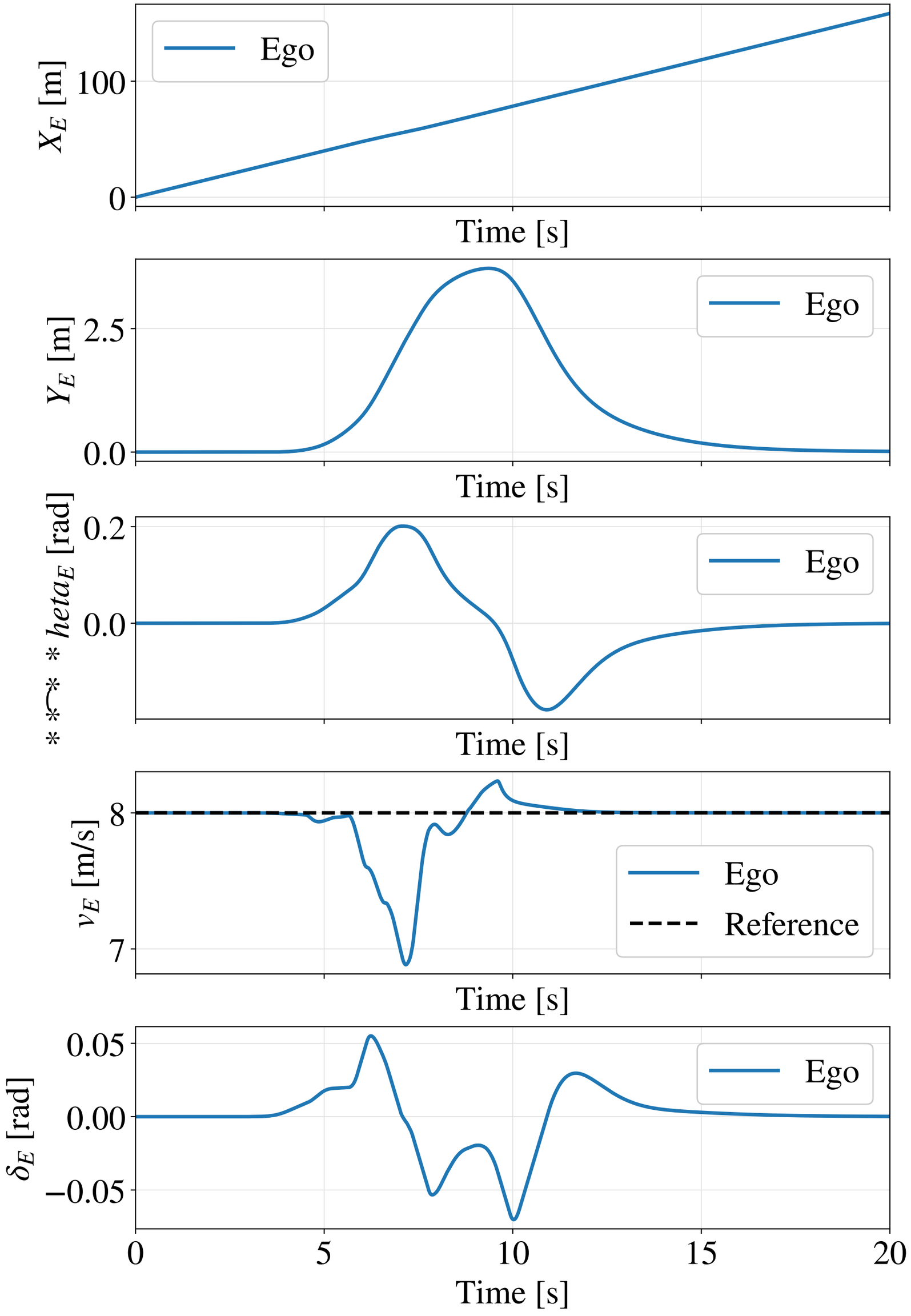
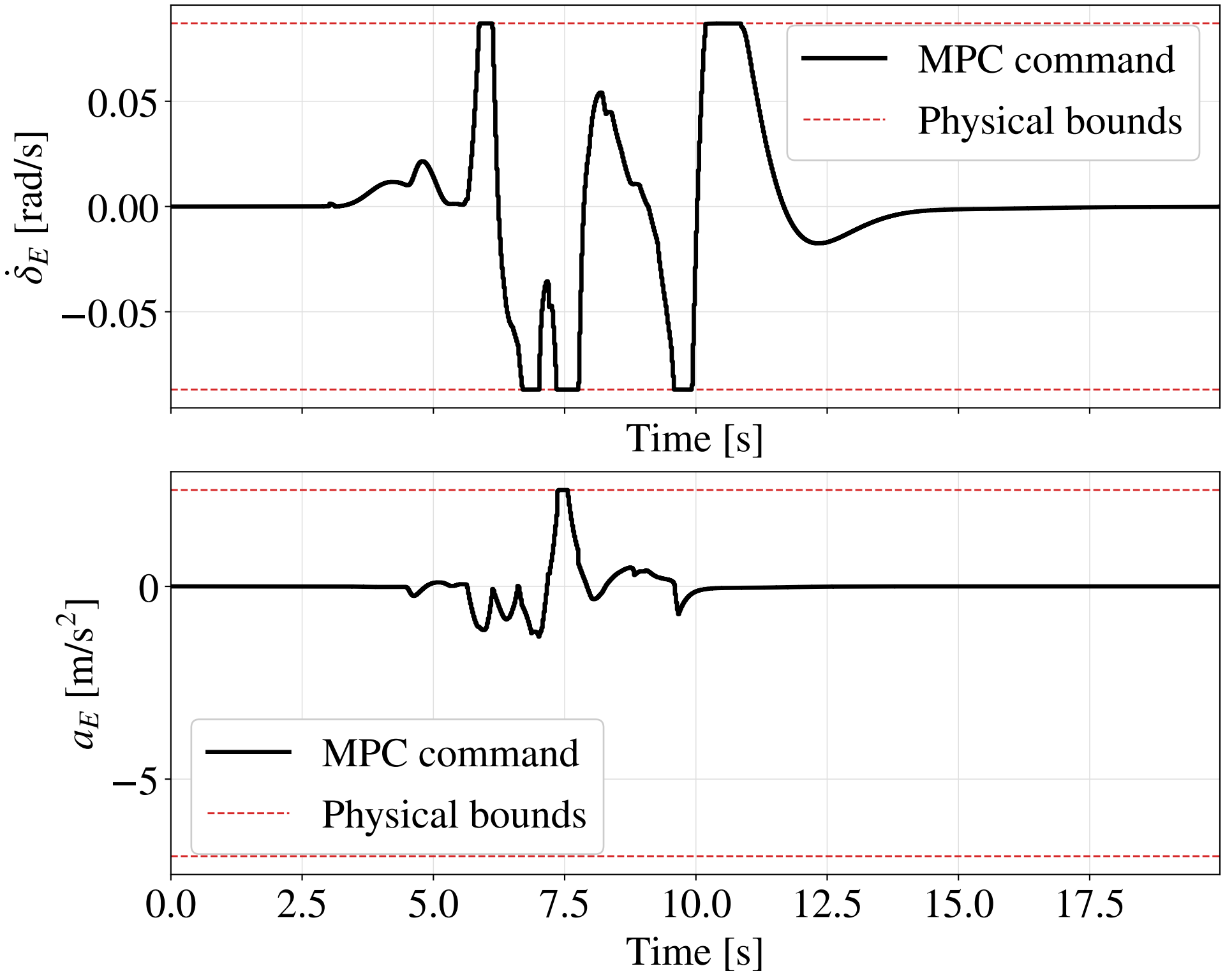
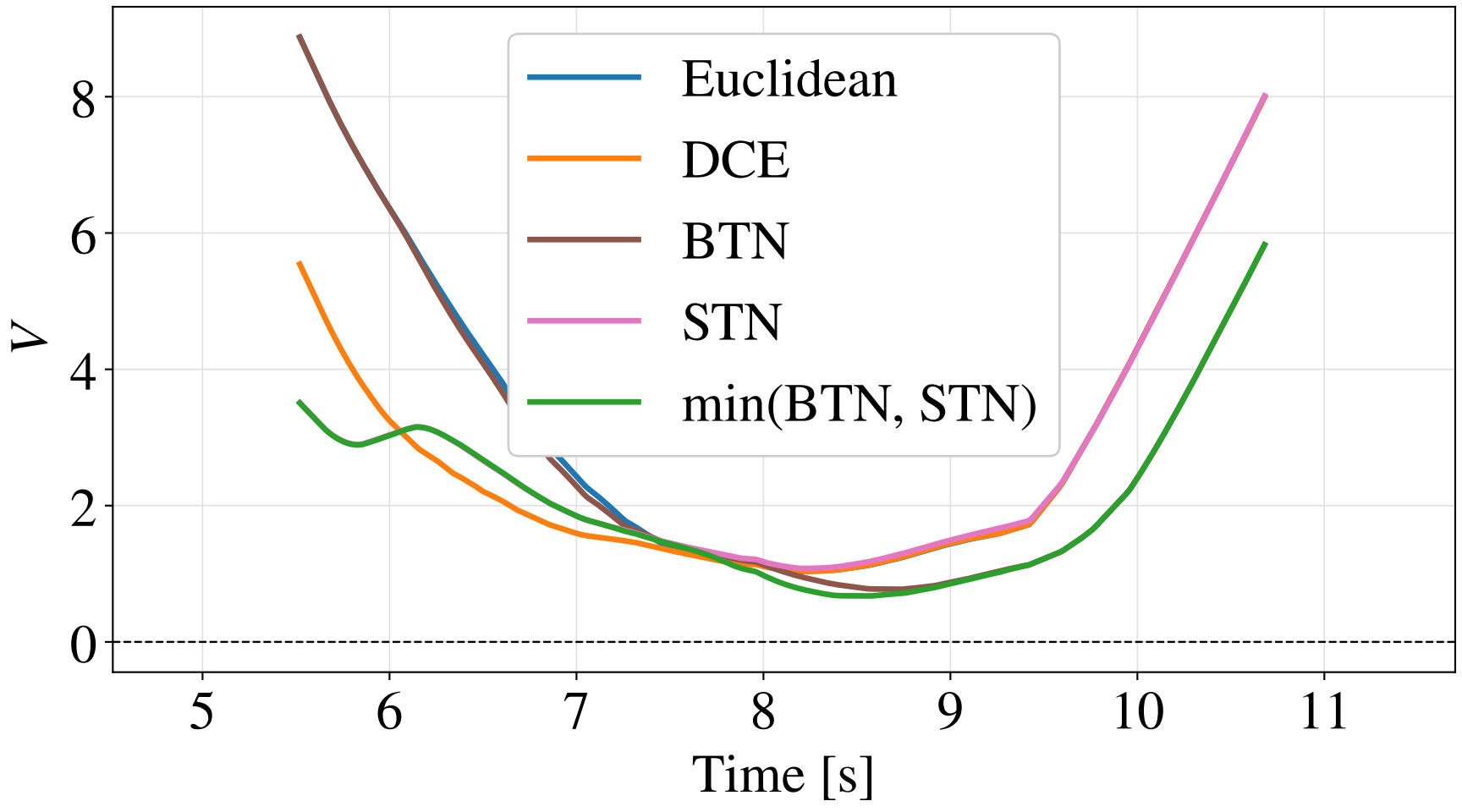
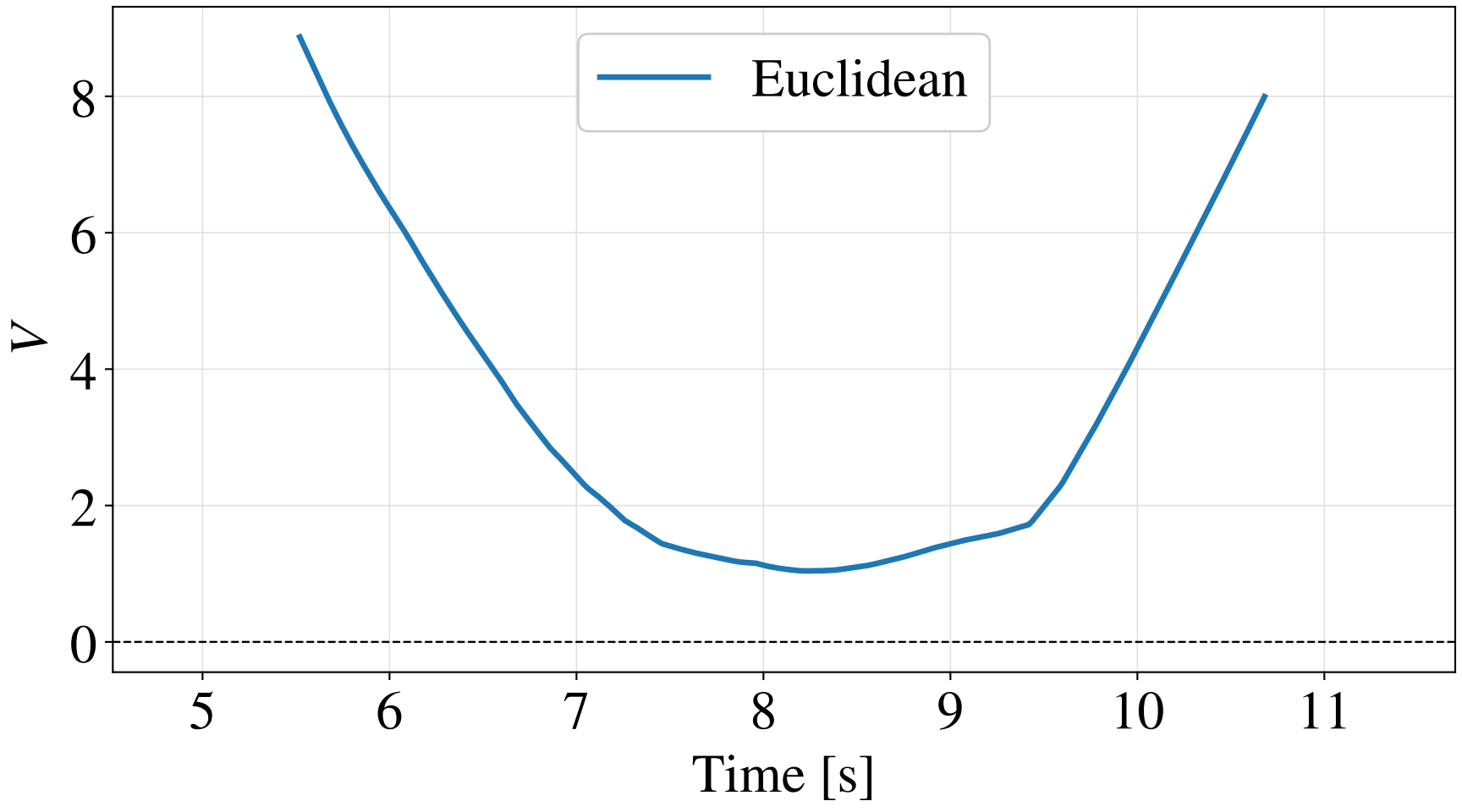
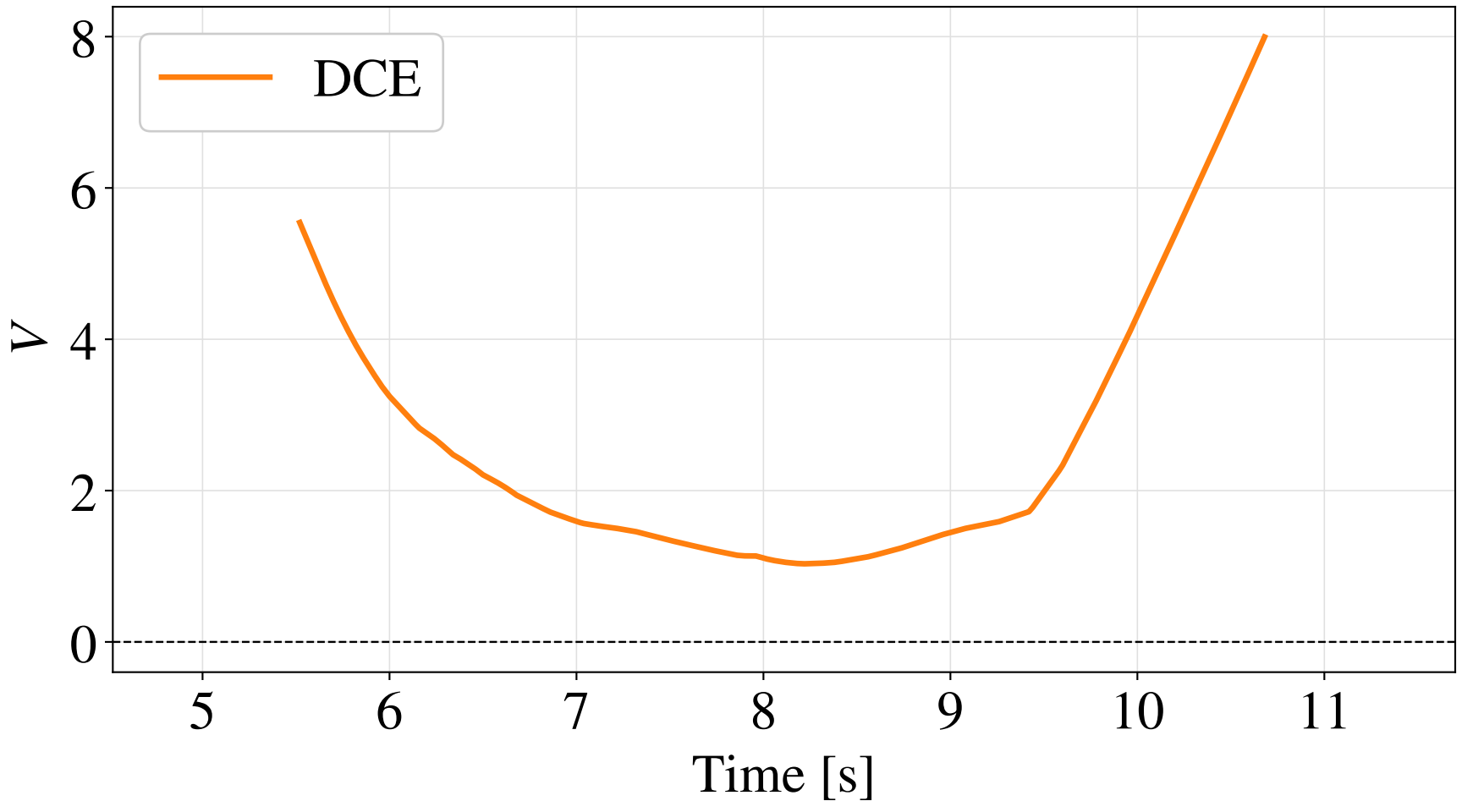

4
3
2
1
0
Y [m]
0
25
50
75
100
125
150
X [m]
20
15
10
5
0
Time [s]

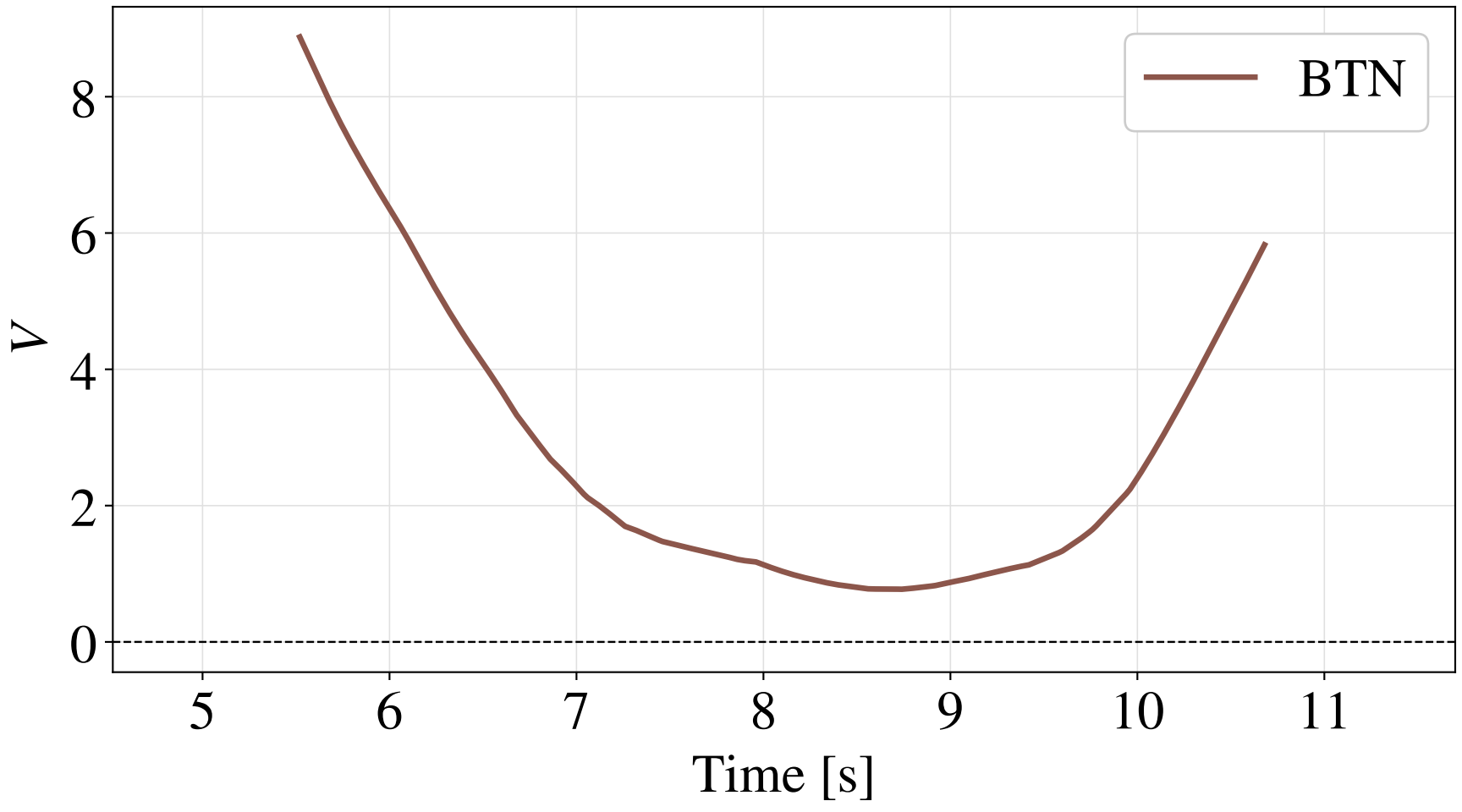

Ego
X_E [m]
100
0
Time [s]
Ego
Y_E [m]
2.5
0.0
Time [s]
Ego
$***** heta_E$ [rad]
0.2
0.0
Time [s]
Ego
Reference
v_E [m/s]
8
7
Time [s]
Ego
δ_E [rad]
0.05
0.00
−0.05
0
5
10
15
20
Time [s]

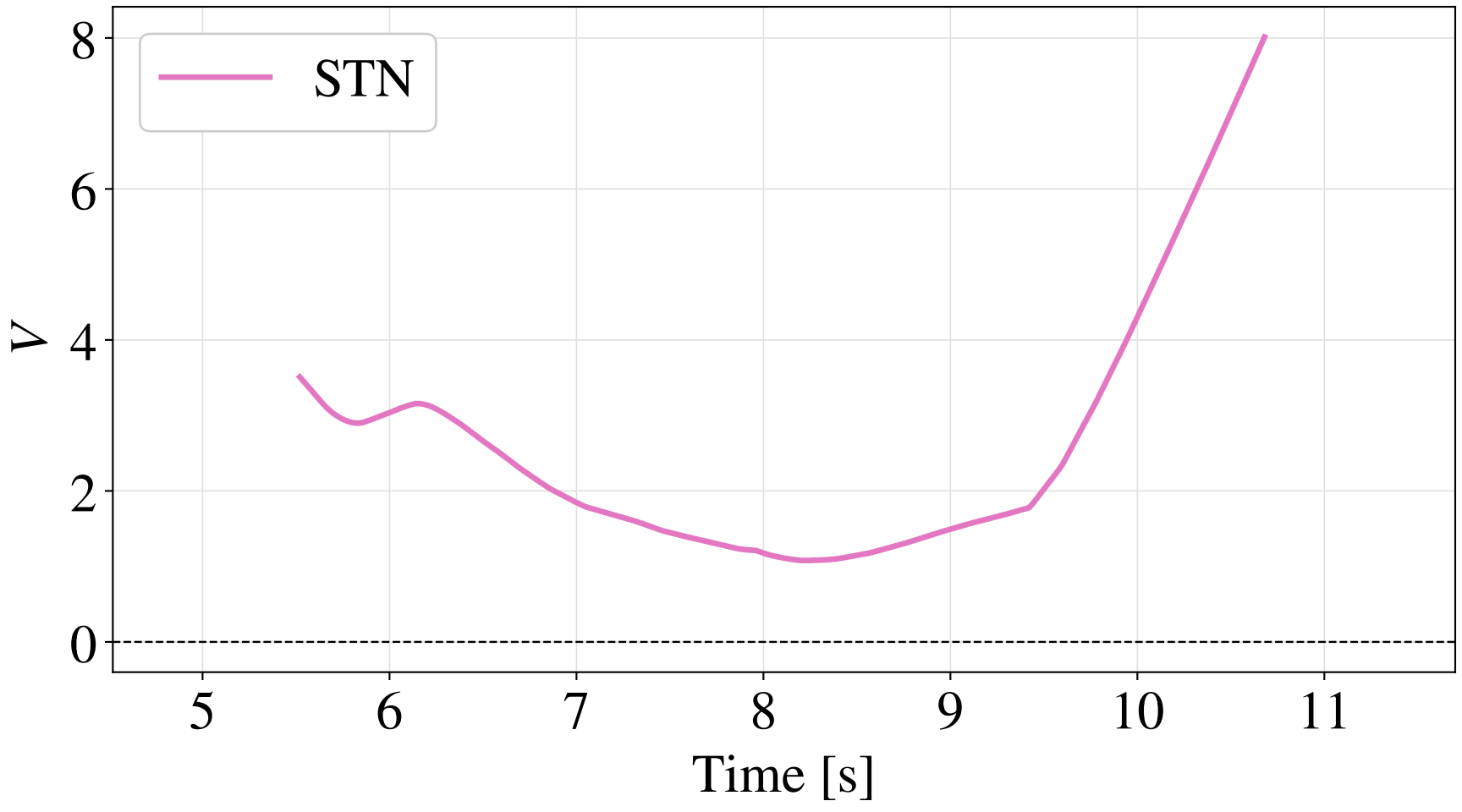

MPC command
Physical bounds
$\dot{\delta}_E$ [rad/s]
0.05
0.00
−0.05
Time [s]
a_E [m/s^2]
0
−5
MPC command
Physical bounds
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]

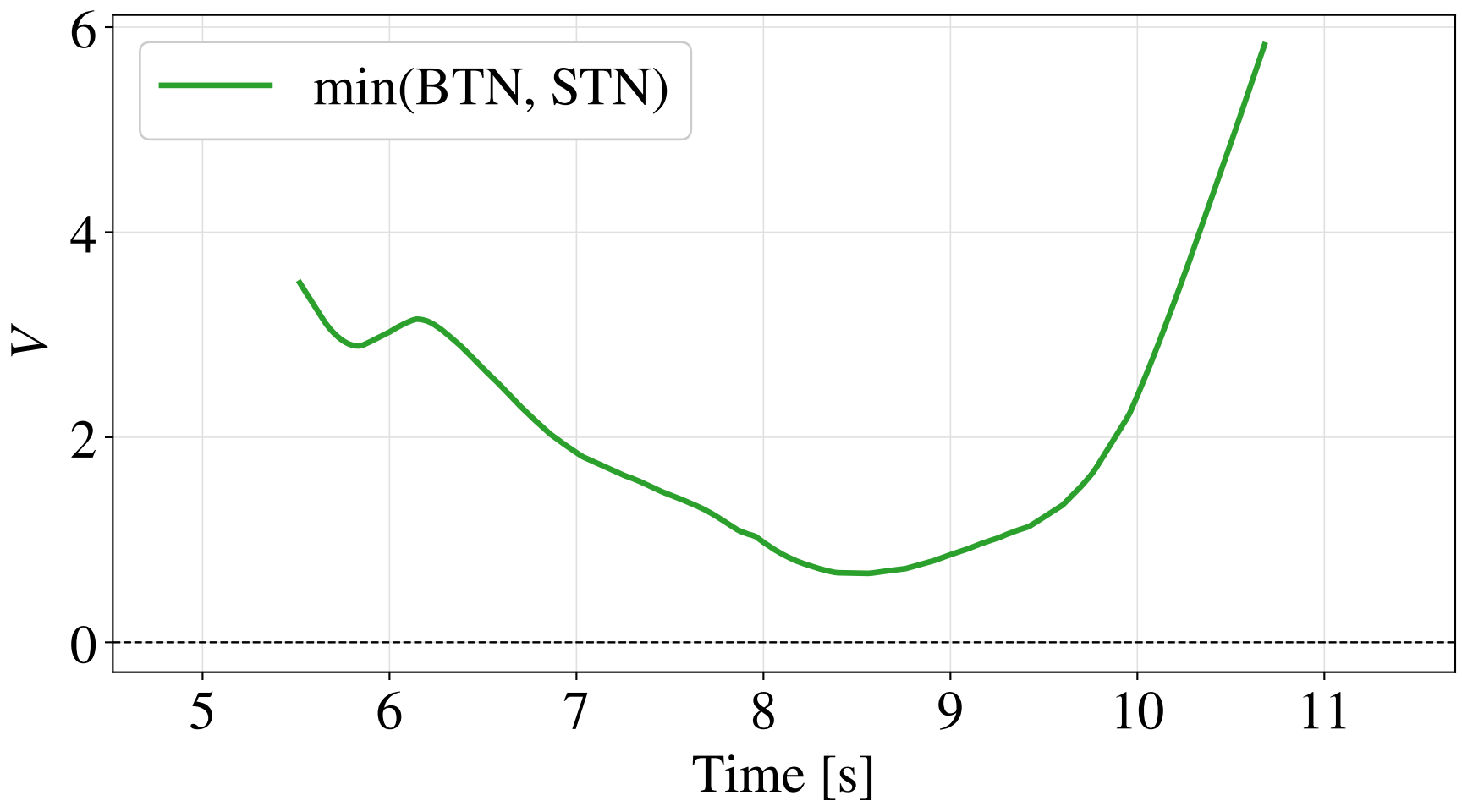

Euclidean
DCE
BTN
STN
min(BTN, STN)
V
0
2
4
6
8
5
6
7
8
9
10
11
Time [s]

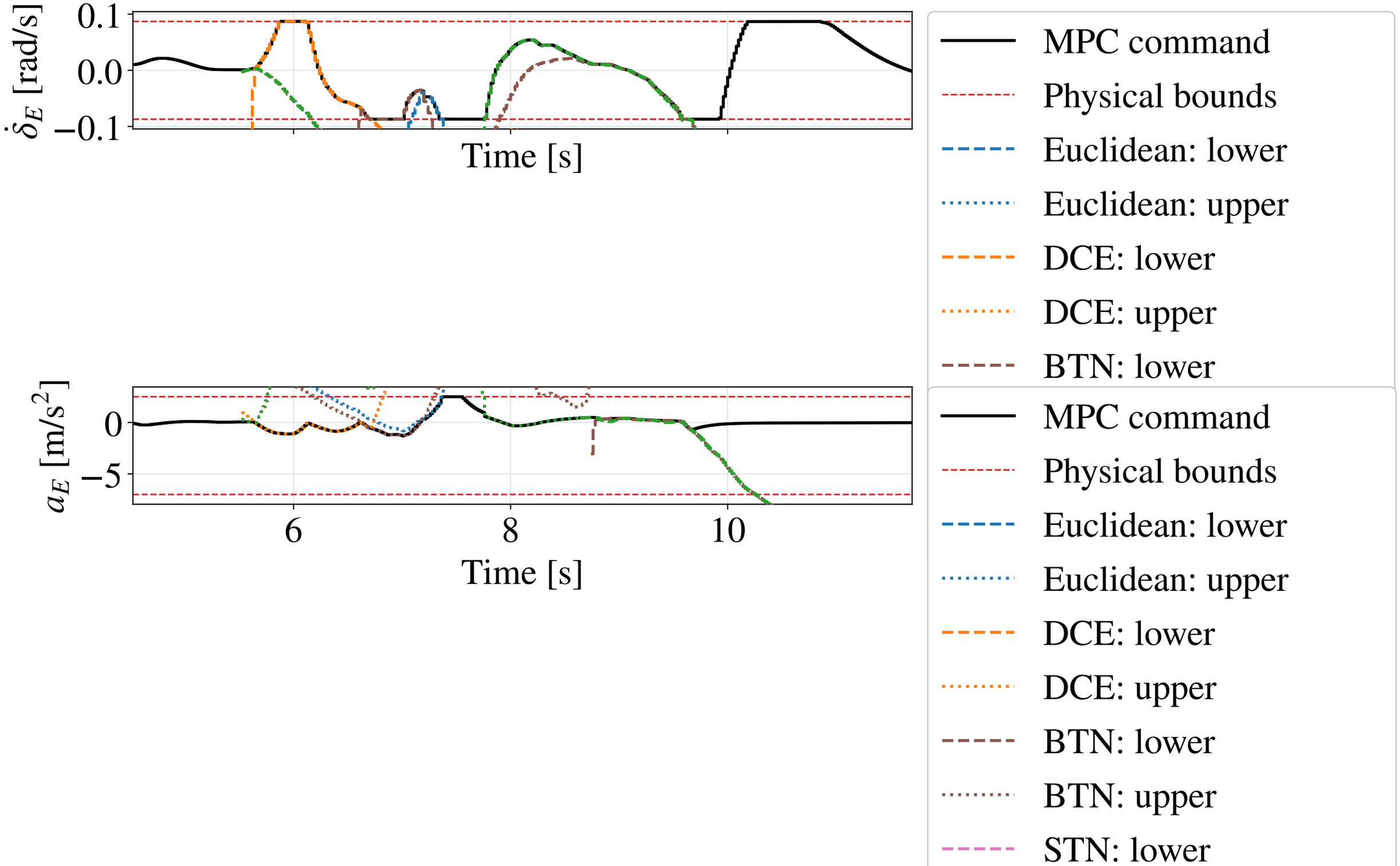

Euclidean
8
6
4
2
0
V
5
6
7
8
9
10
11
Time [s]

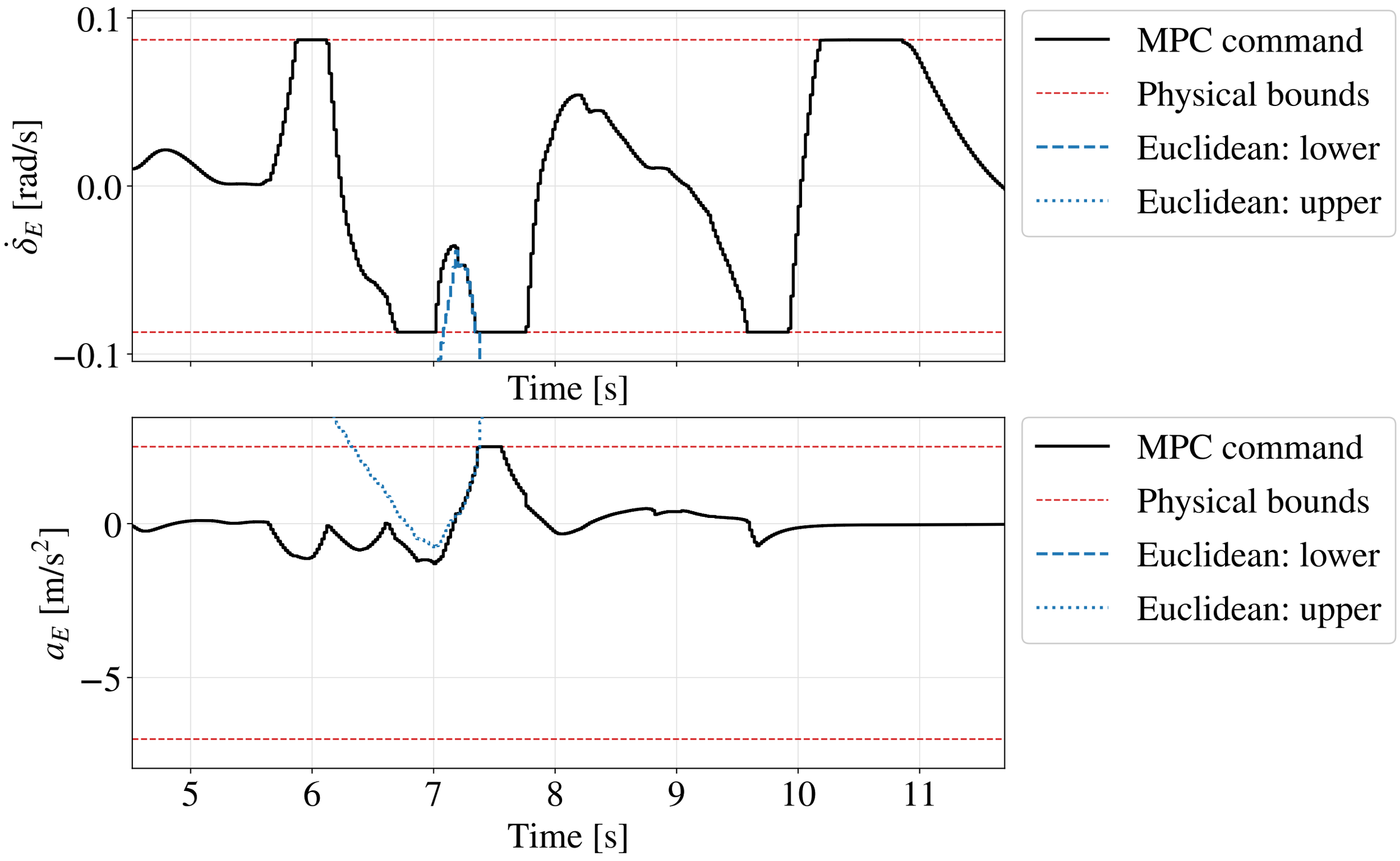

DCE
8
6
4
2
0
V
5
6
7
8
9
10
11
Time [s]

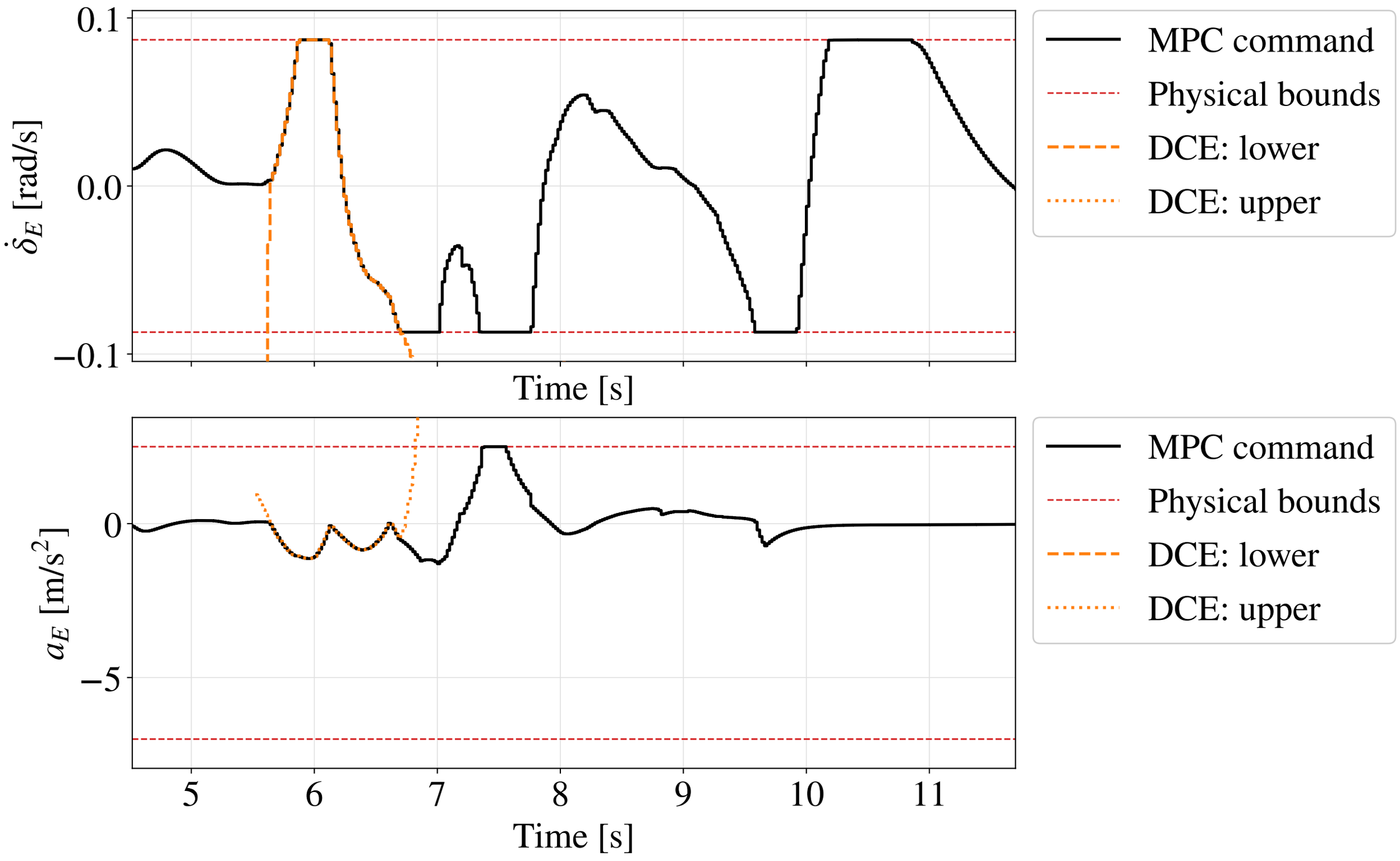

BTN
8
6
4
2
0
V
5
6
7
8
9
10
11
Time [s]

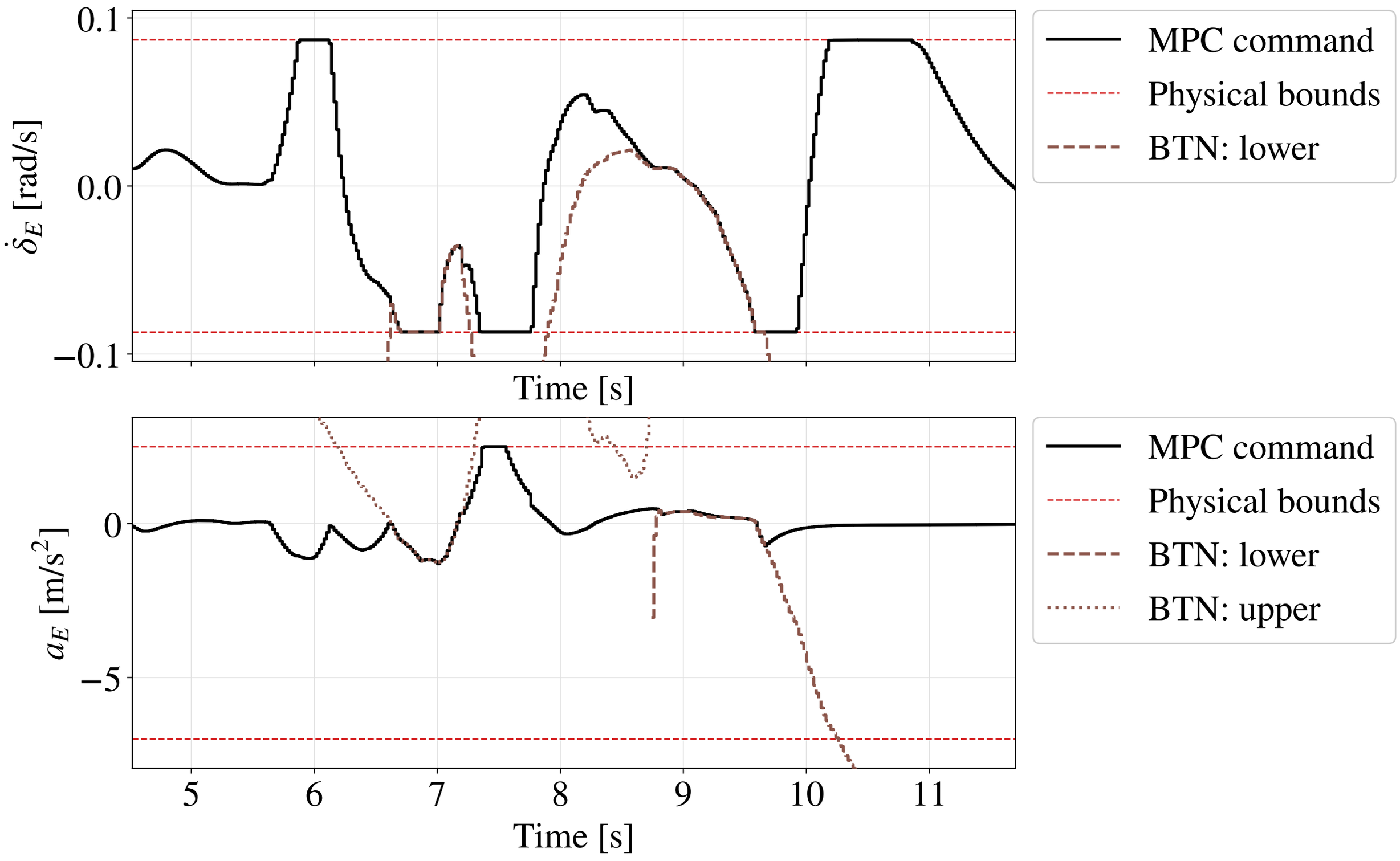

STN
8
6
4
2
0
V
5
6
7
8
9
10
11
Time [s]

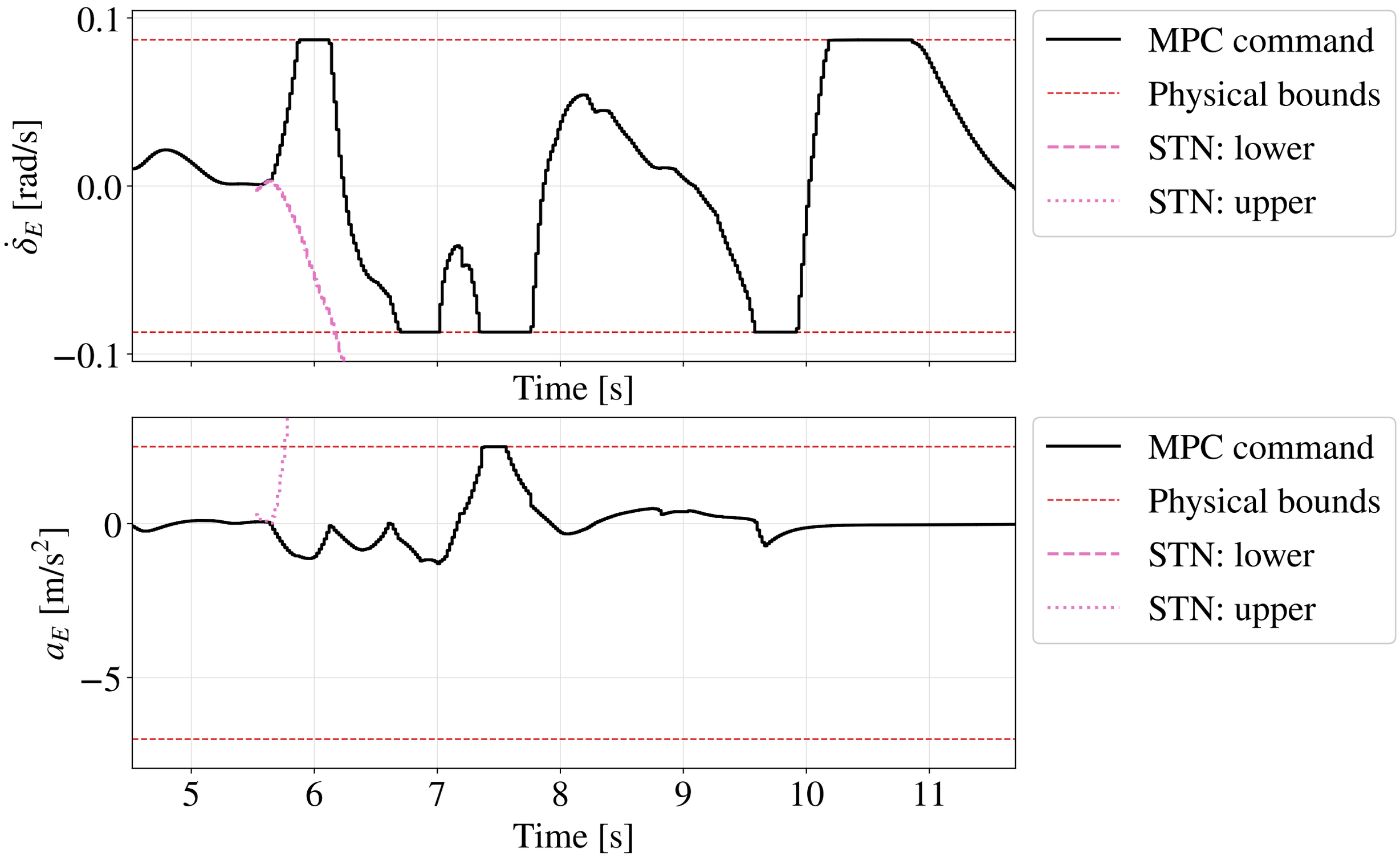

min(BTN, STN)
V
6
4
2
0
5
6
7
8
9
10
11
Time [s]

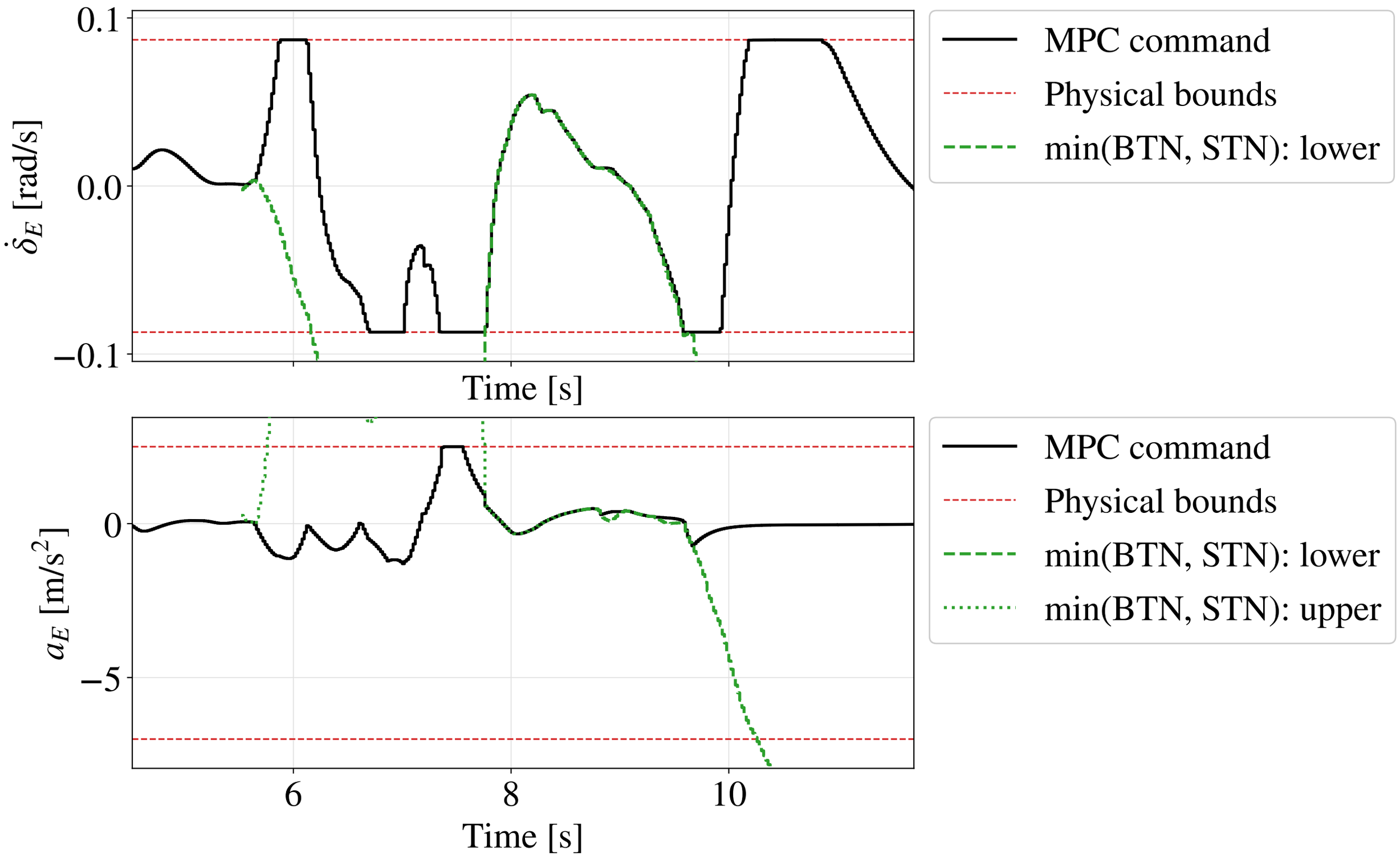

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
0
−5
a_E [m/s^2]
6
8
10
Time [s]
MPC command
Physical bounds
Euclidean: lower
Euclidean: upper
DCE: lower
DCE: upper
BTN: lower
MPC command
Physical bounds
Euclidean: lower
Euclidean: upper
DCE: lower
DCE: upper
BTN: lower
BTN: upper
STN: lower

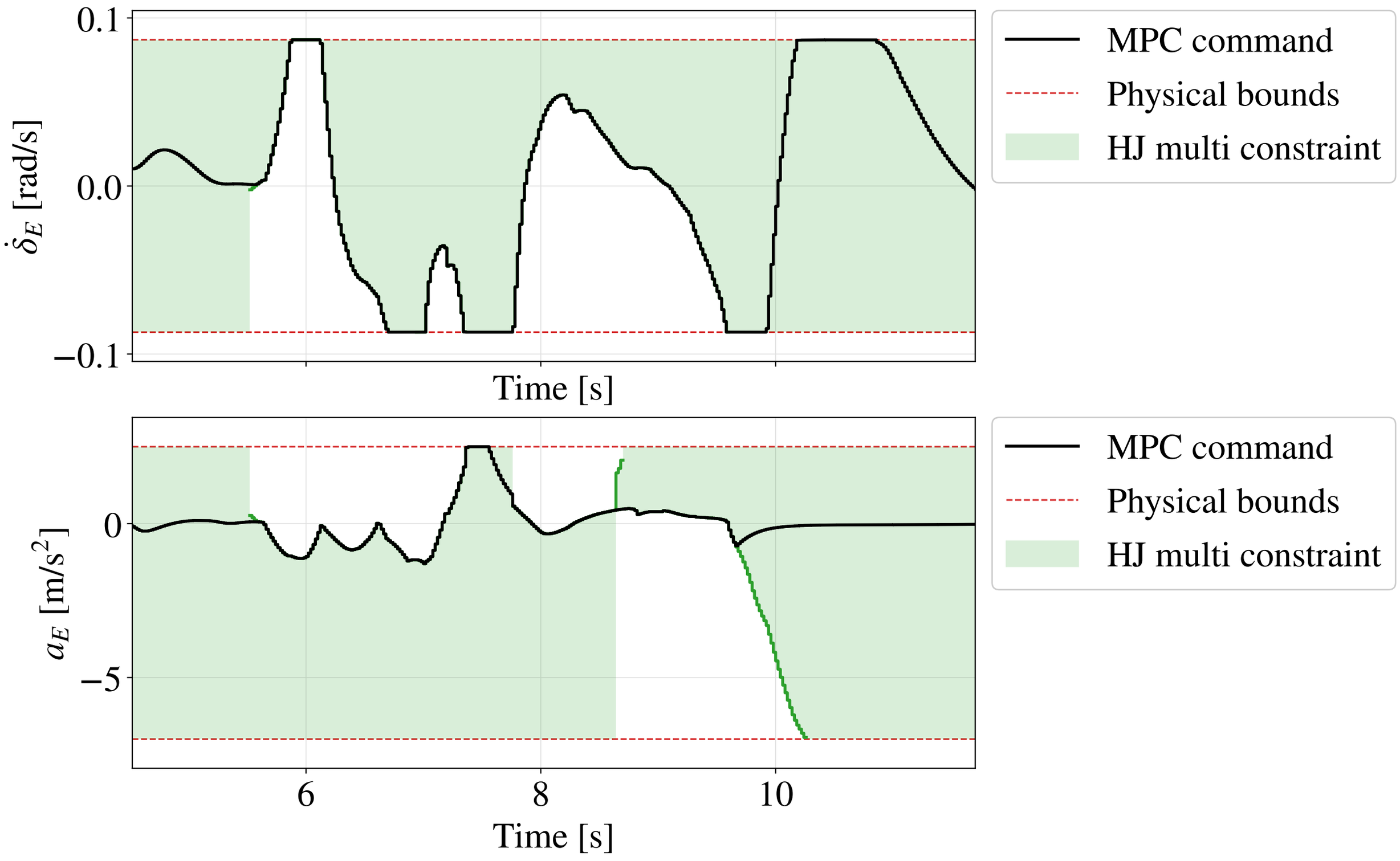

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
Euclidean: lower
Euclidean: upper
0
−5
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
Euclidean: lower
Euclidean: upper

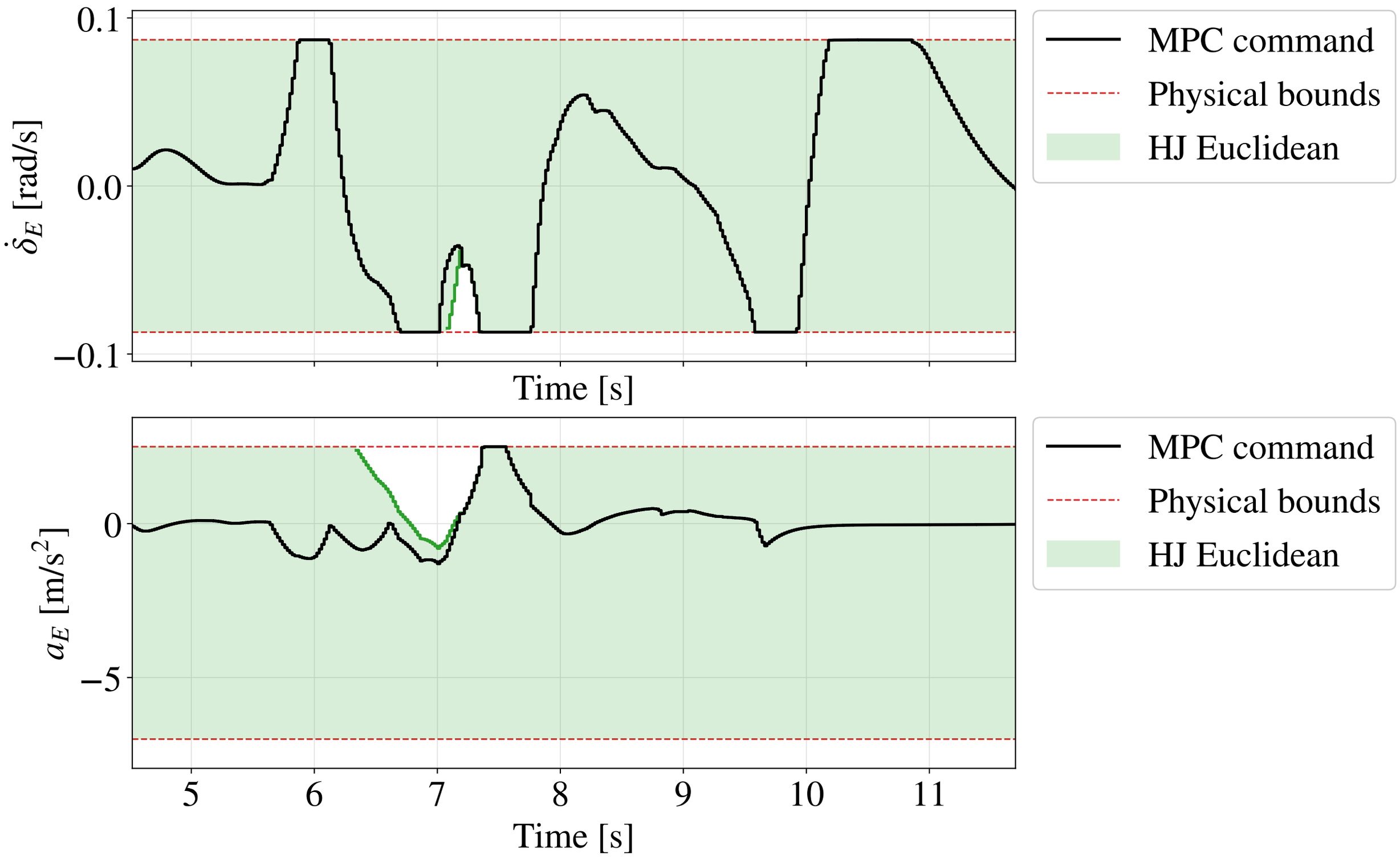

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
DCE: lower
DCE: upper
0
−5
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
DCE: lower
DCE: upper

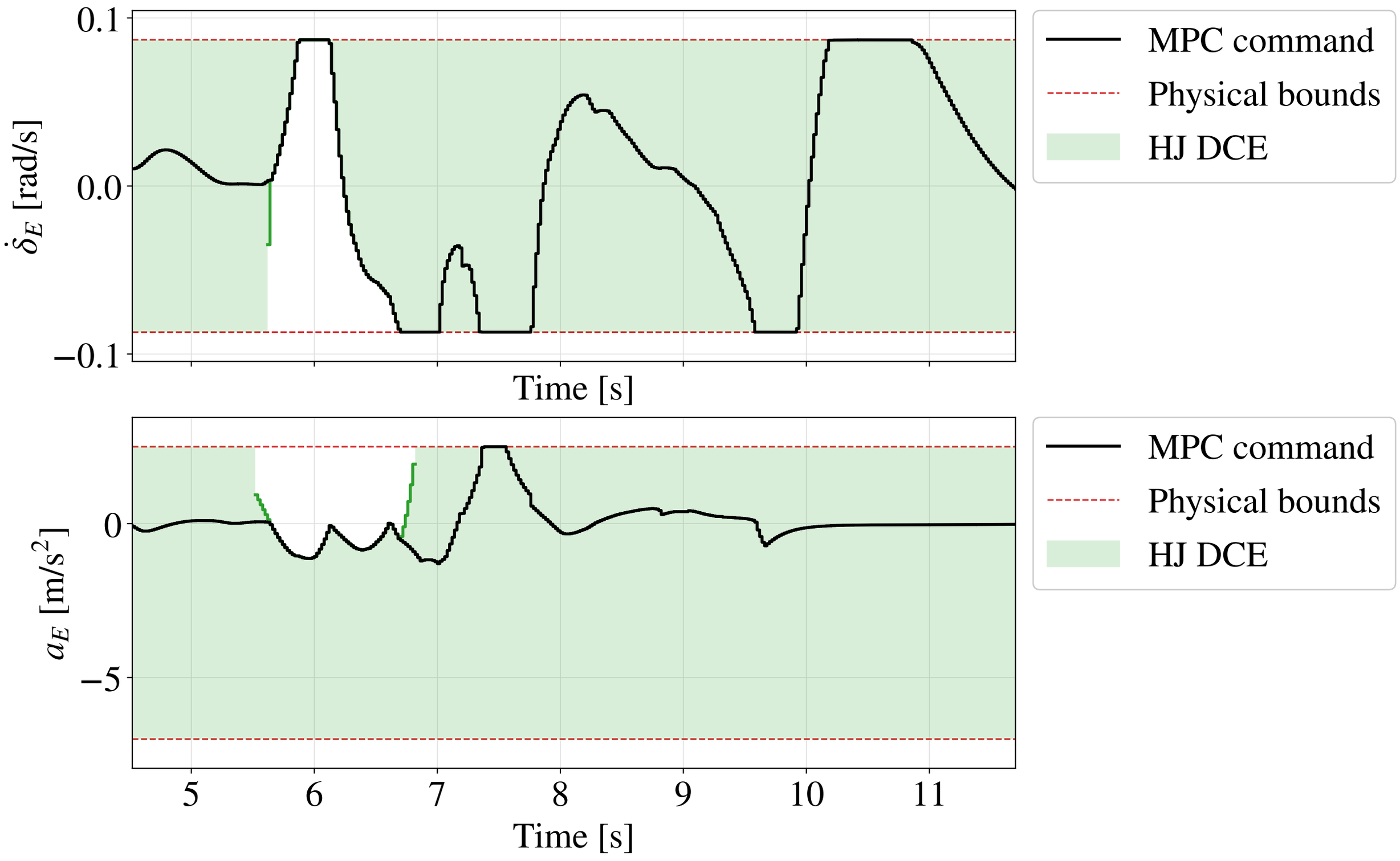

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
BTN: lower
0
−5
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
BTN: lower
BTN: upper

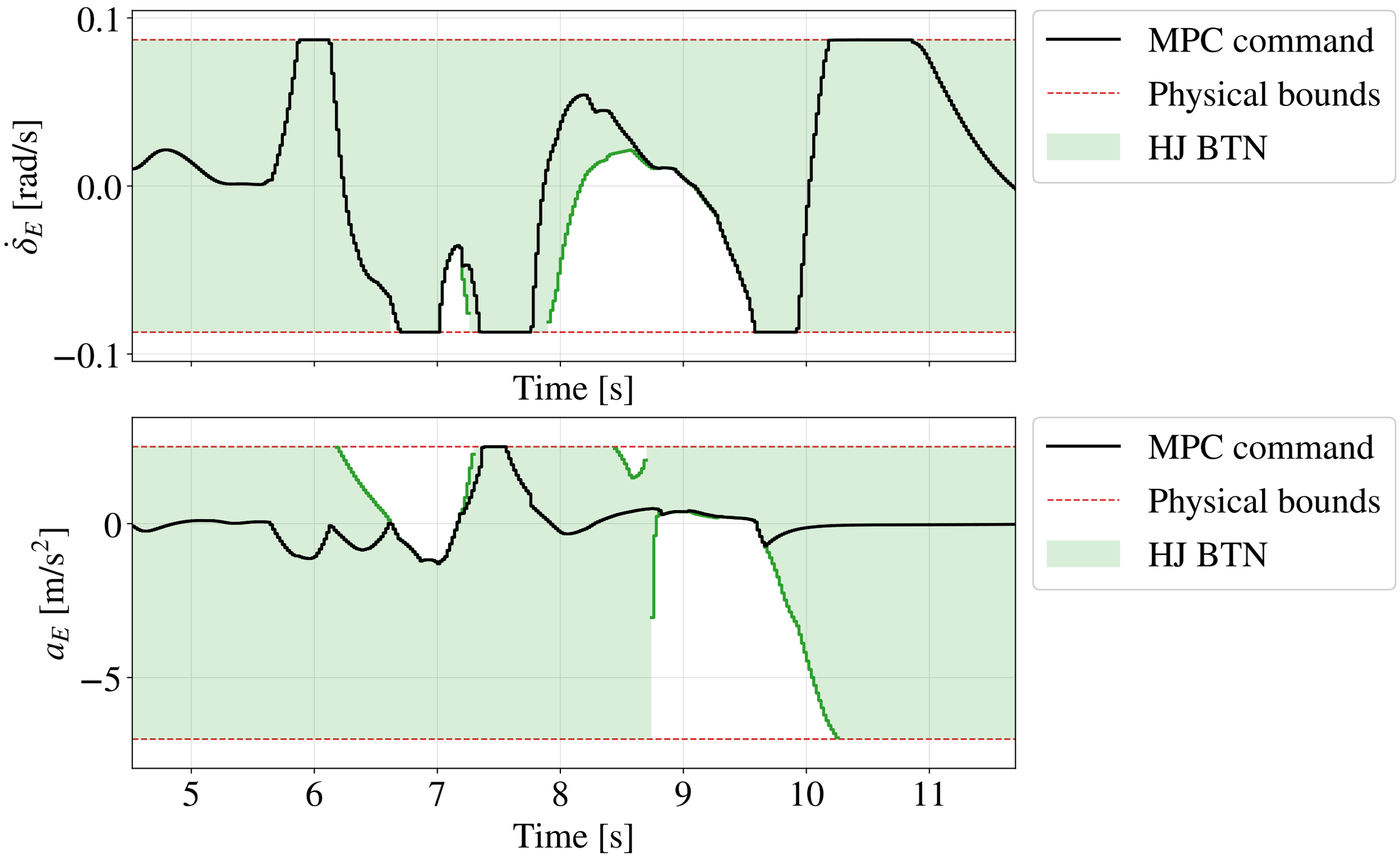

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
STN: lower
STN: upper
0
−5
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
STN: lower
STN: upper

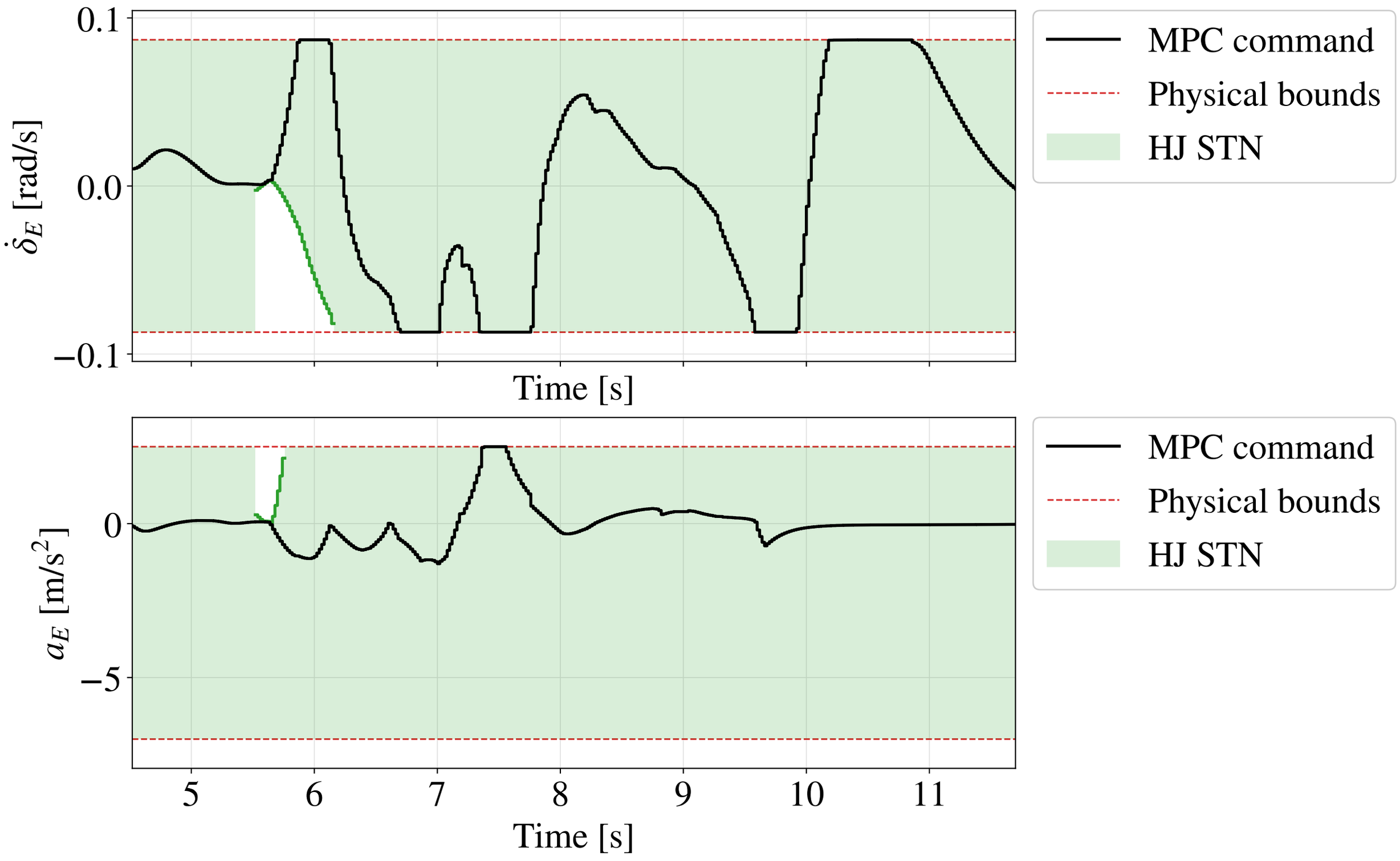

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
min(BTN, STN): lower
0
−5
a_E [m/s^2]
6
8
10
Time [s]
MPC command
Physical bounds
min(BTN, STN): lower
min(BTN, STN): upper

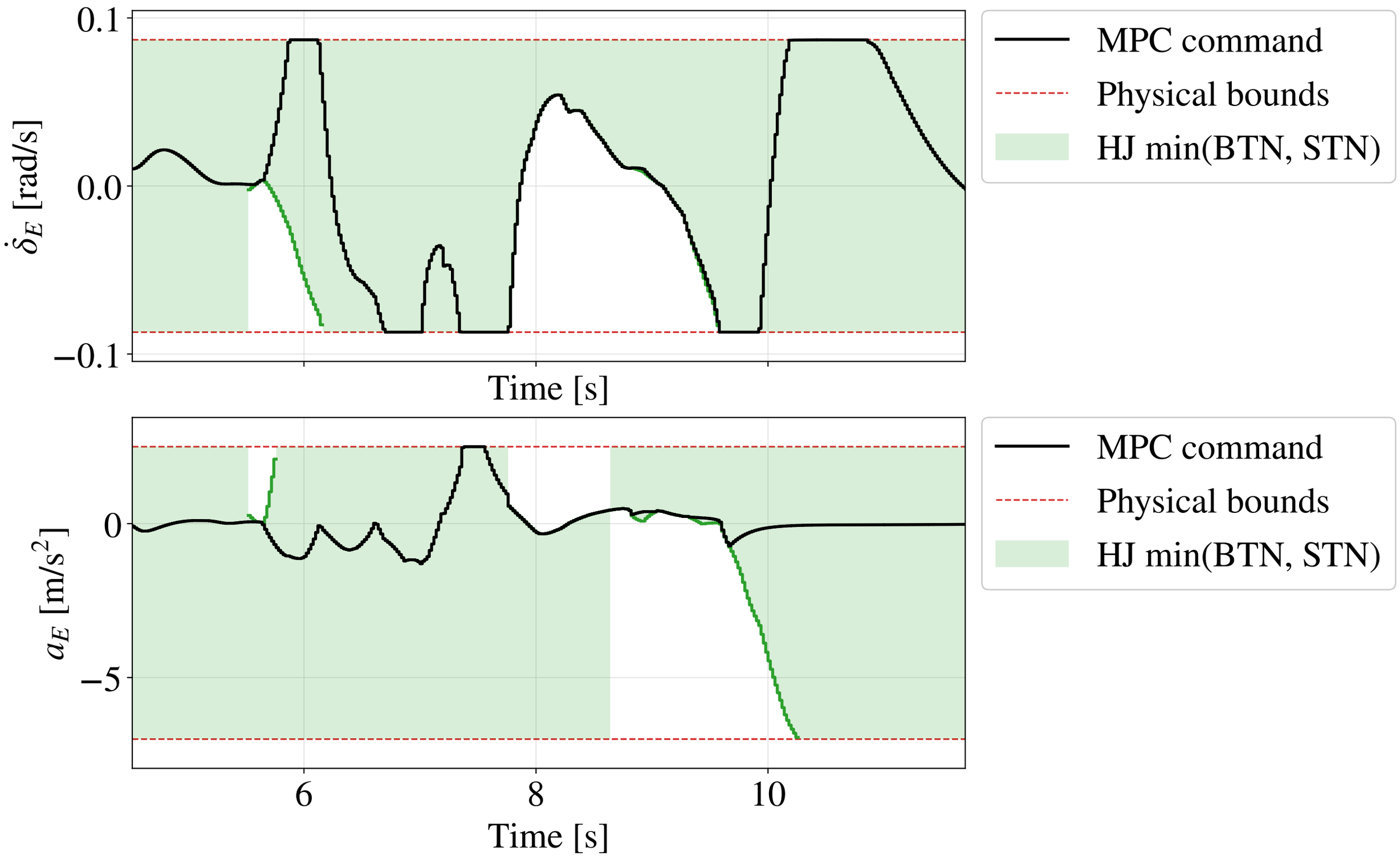

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
HJ multi constraint
0
−5
a_E [m/s^2]
6
8
10
Time [s]
MPC command
Physical bounds
HJ multi constraint

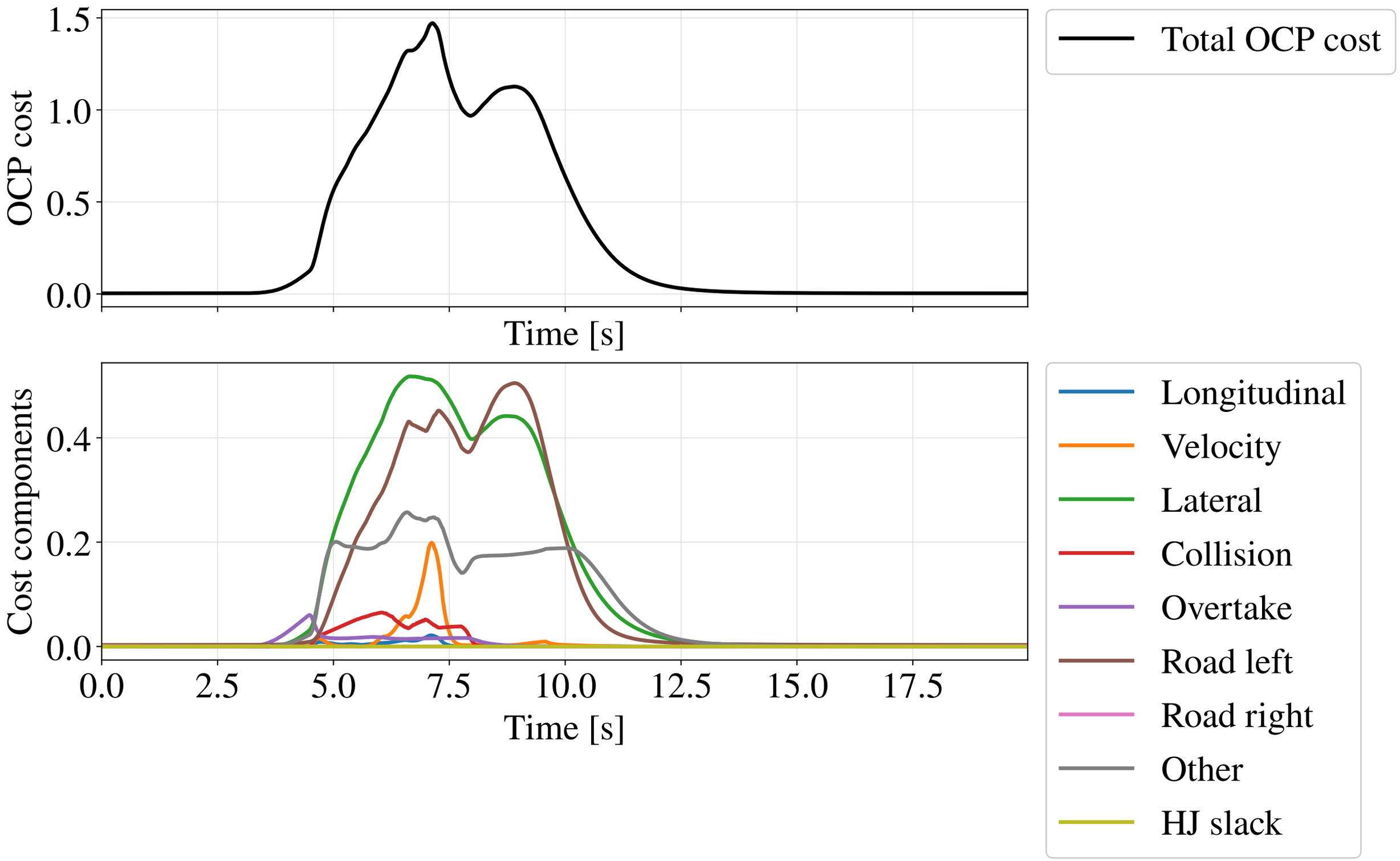

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
HJ Euclidean
0
−5
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
HJ Euclidean

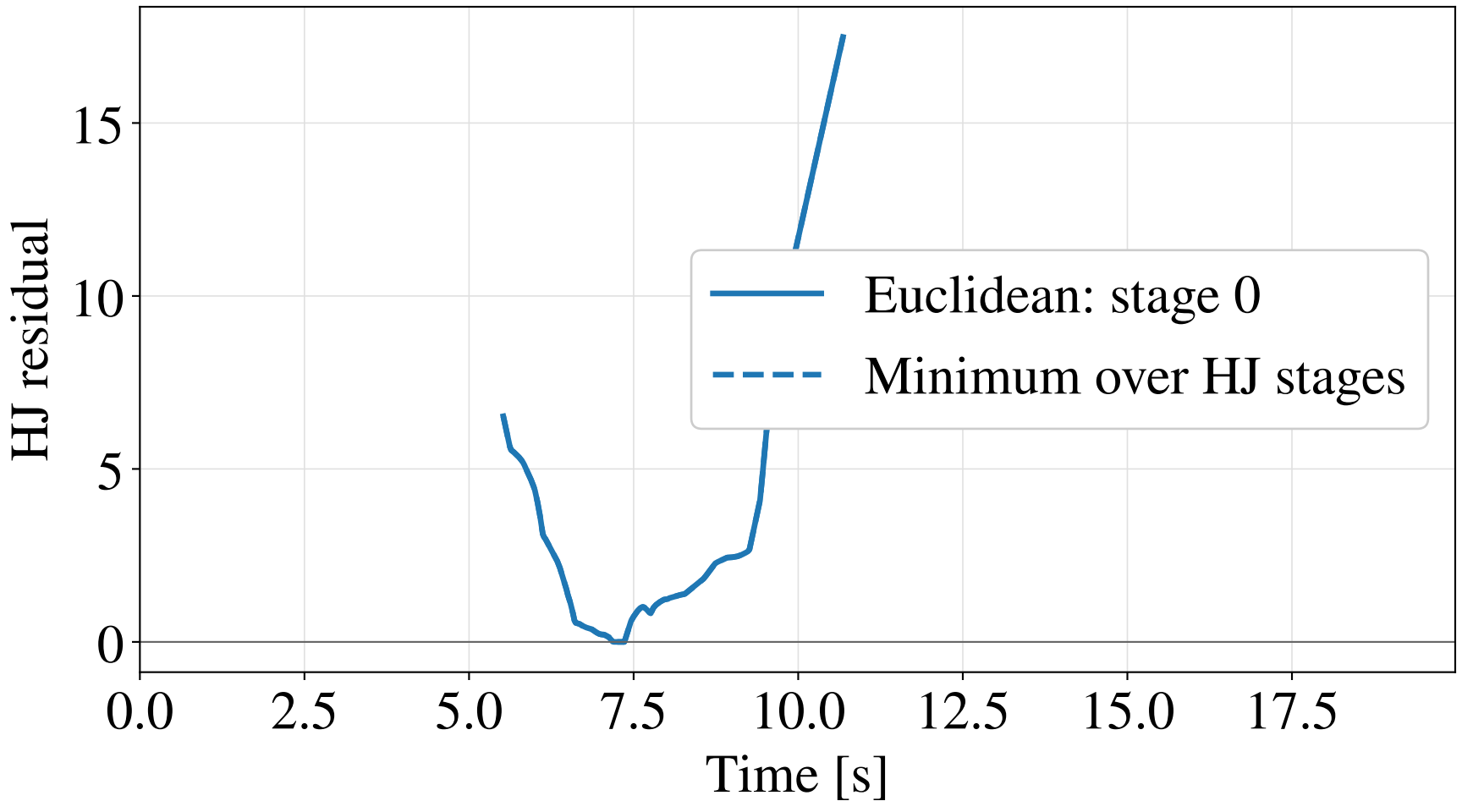

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
HJ DCE
0
−5
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
HJ DCE

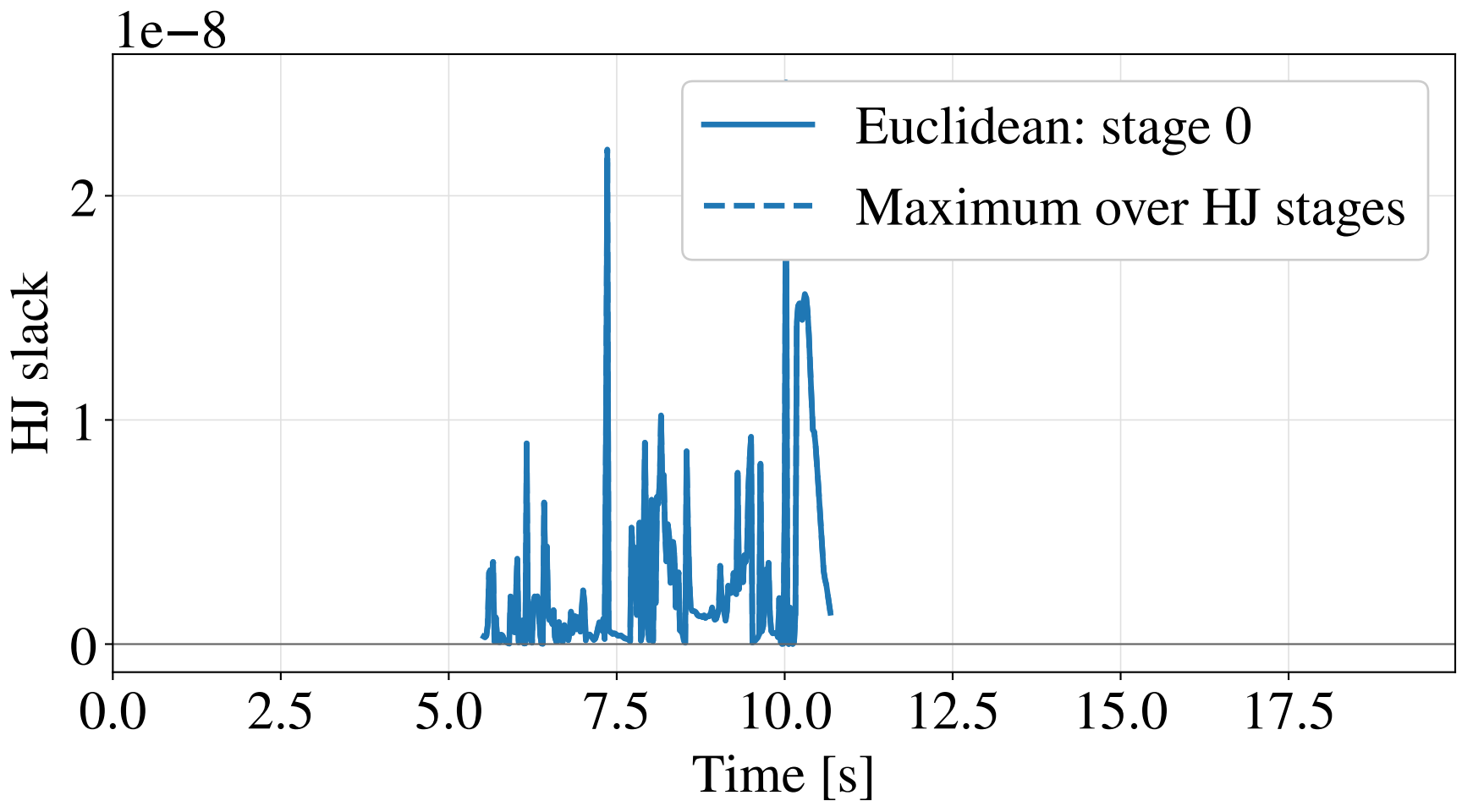

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
HJ BTN
0
−5
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
HJ BTN

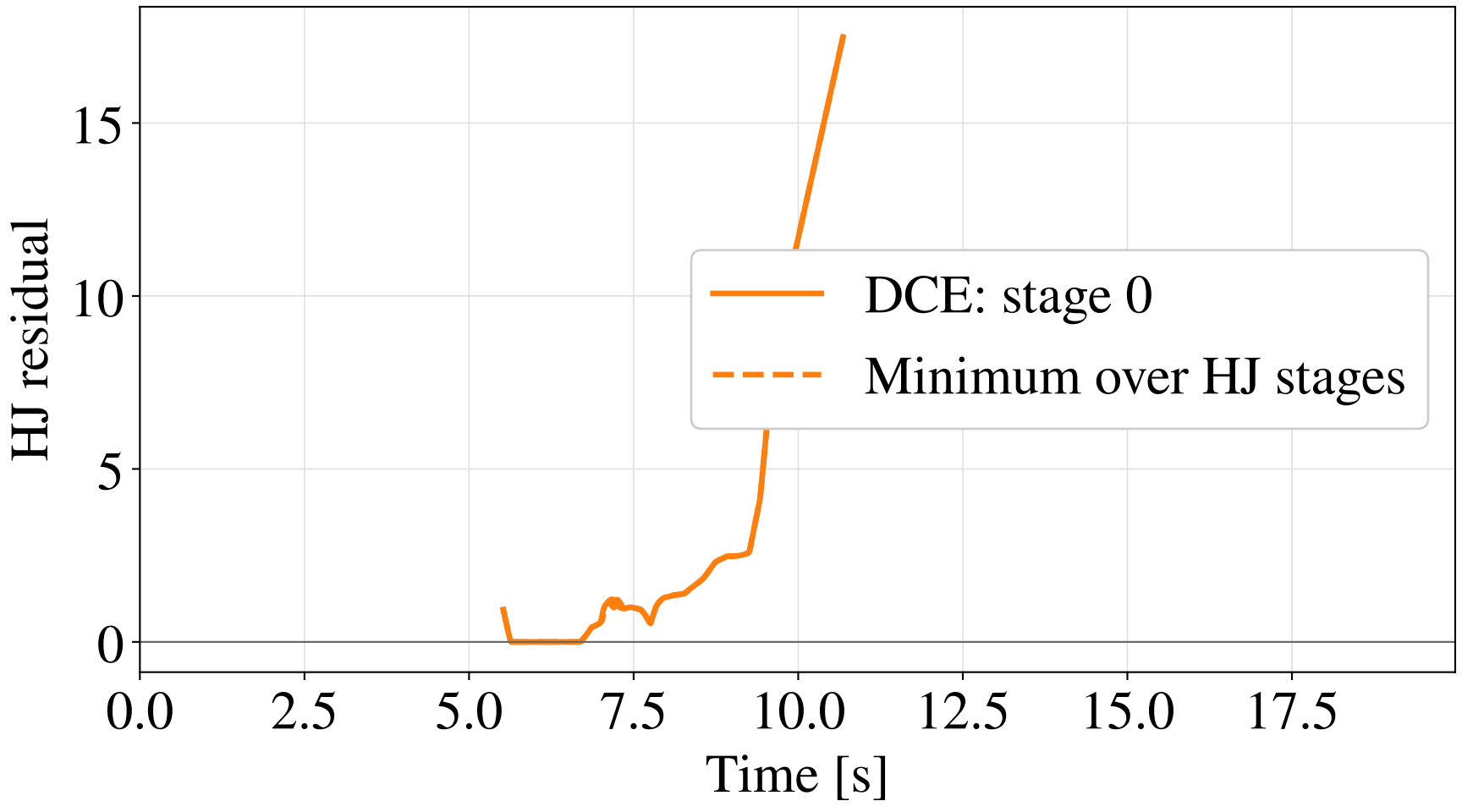

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
HJ STN
0
−5
a_E [m/s^2]
5
6
7
8
9
10
11
Time [s]
MPC command
Physical bounds
HJ STN

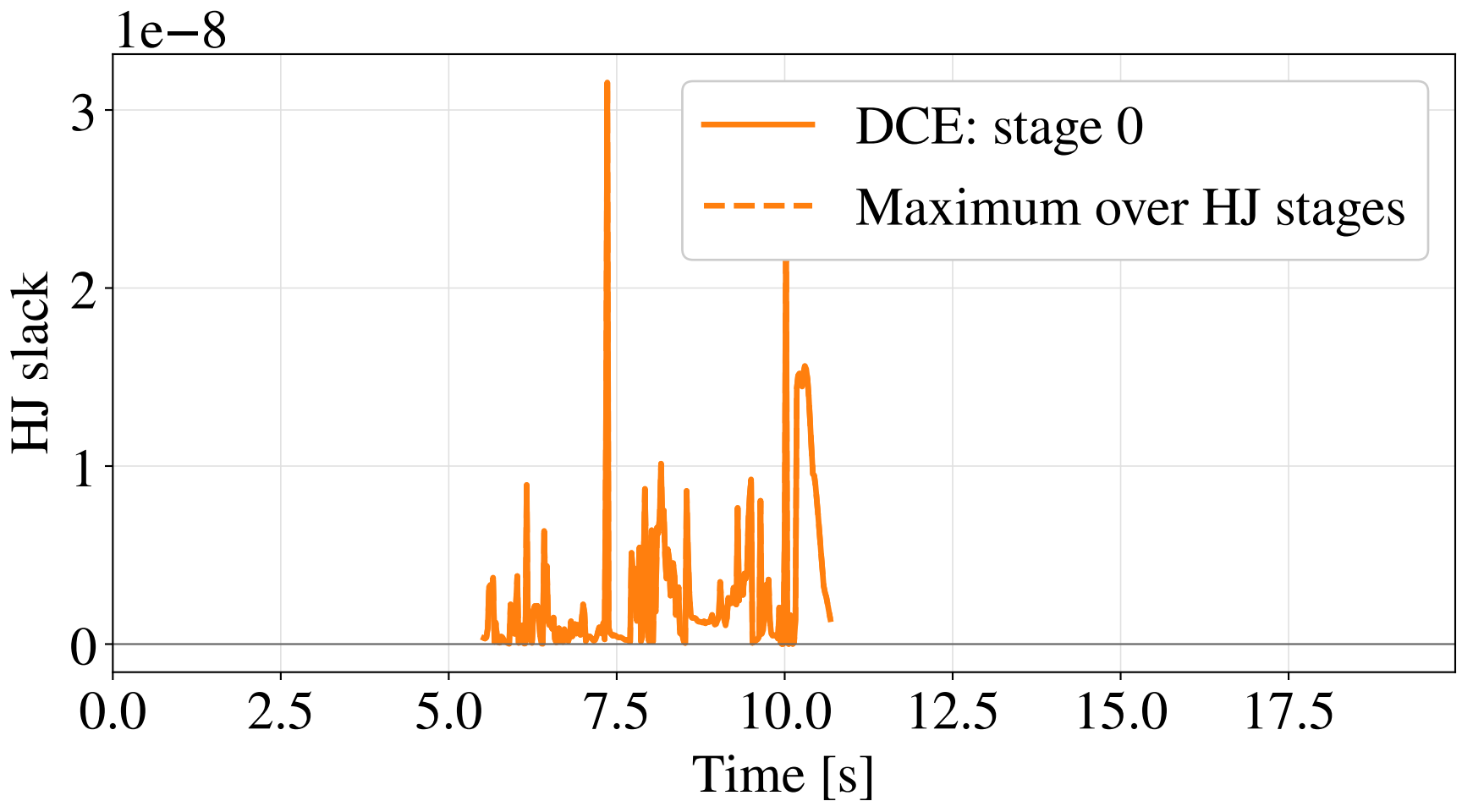

0.1
0.0
−0.1
$\dot{\delta}_E$ [rad/s]
Time [s]
MPC command
Physical bounds
HJ min(BTN, STN)
0
−5
a_E [m/s^2]
6
8
10
Time [s]
MPC command
Physical bounds
HJ min(BTN, STN)

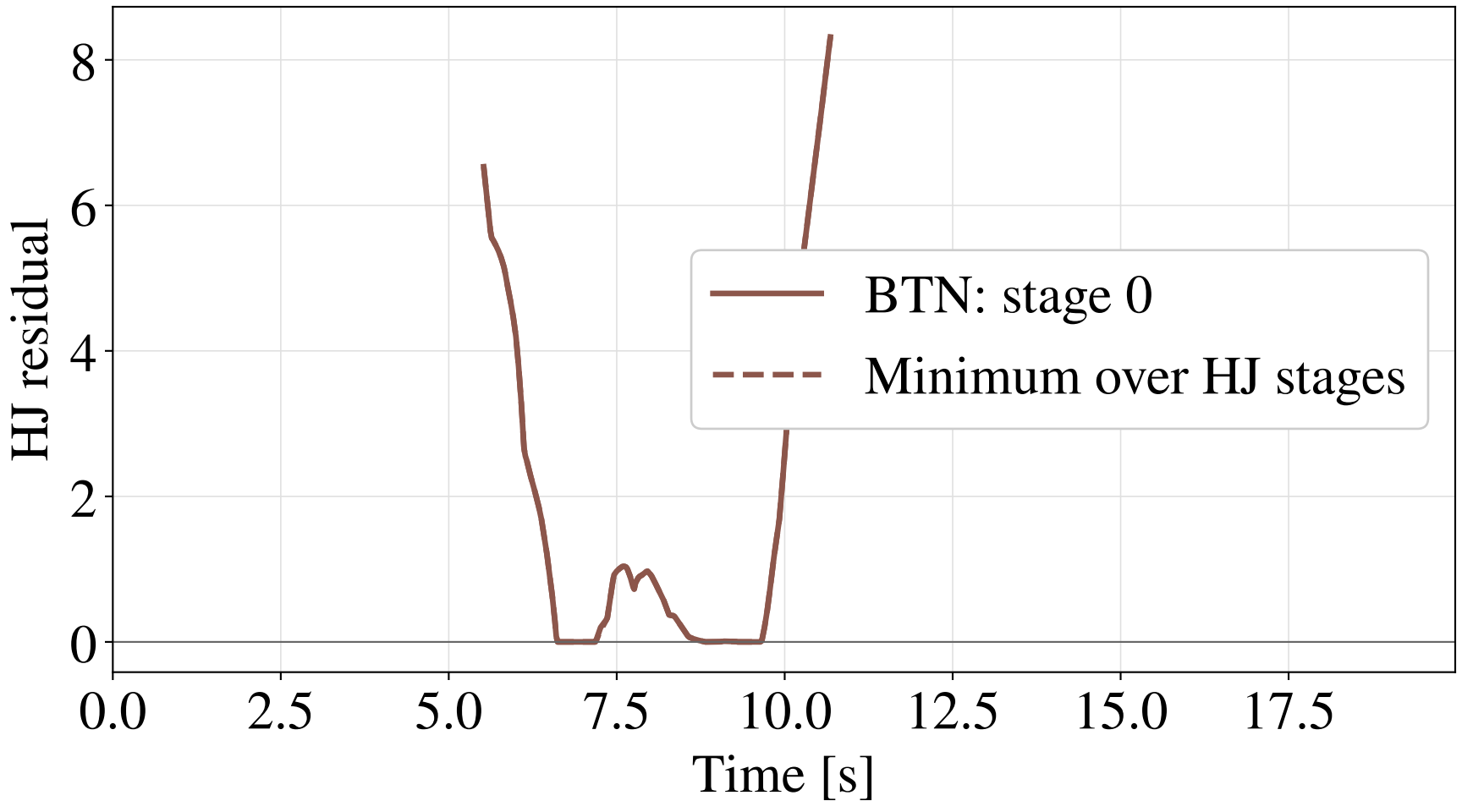

OCP cost
1.5
1.0
0.5
0.0
Time [s]
Total OCP cost
Cost components
0.4
0.2
0.0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]
Longitudinal
Velocity
Lateral
Collision
Overtake
Road left
Road right
Other
HJ slack

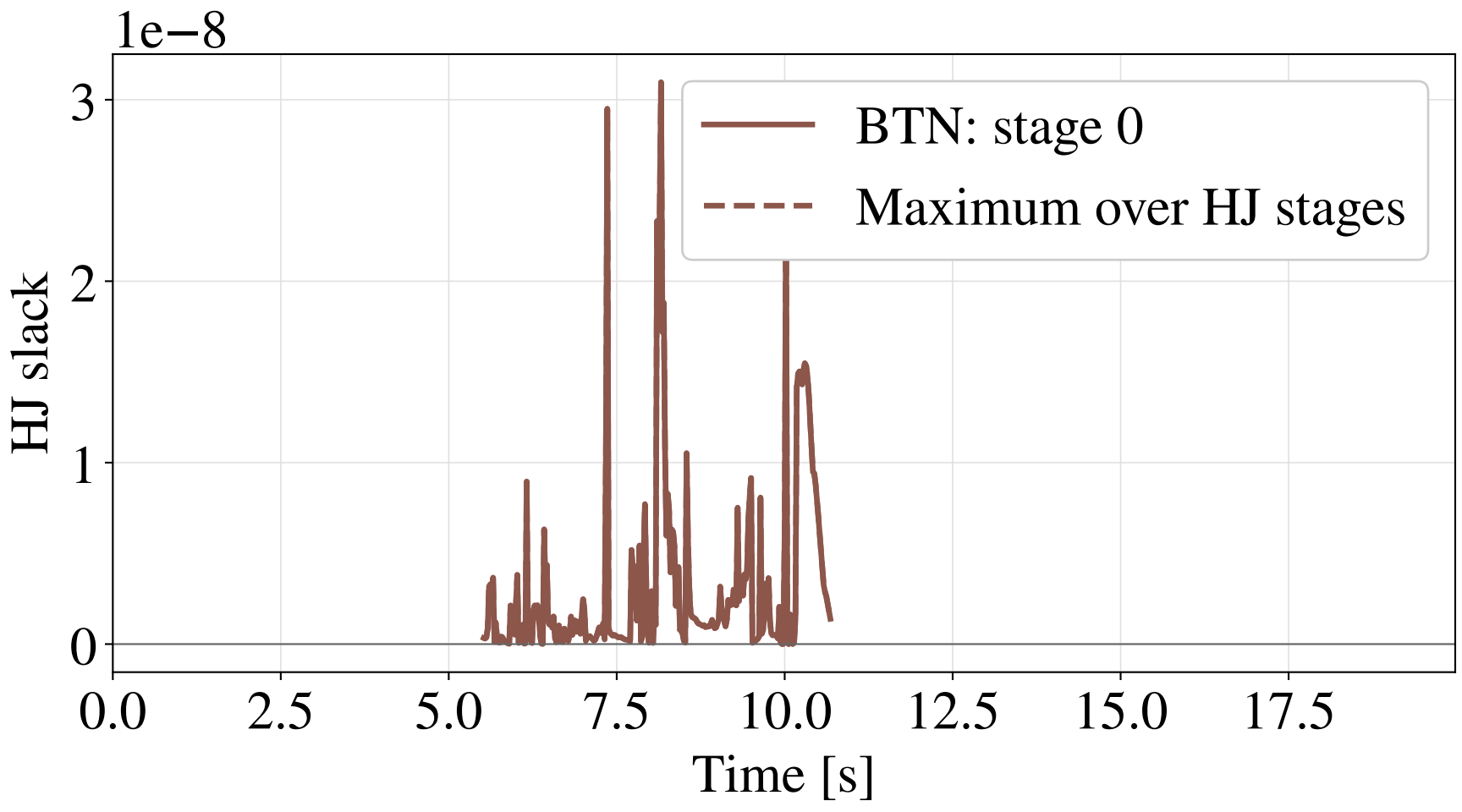

HJ residual
15
10
5
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]
Euclidean: stage 0
Minimum over HJ stages

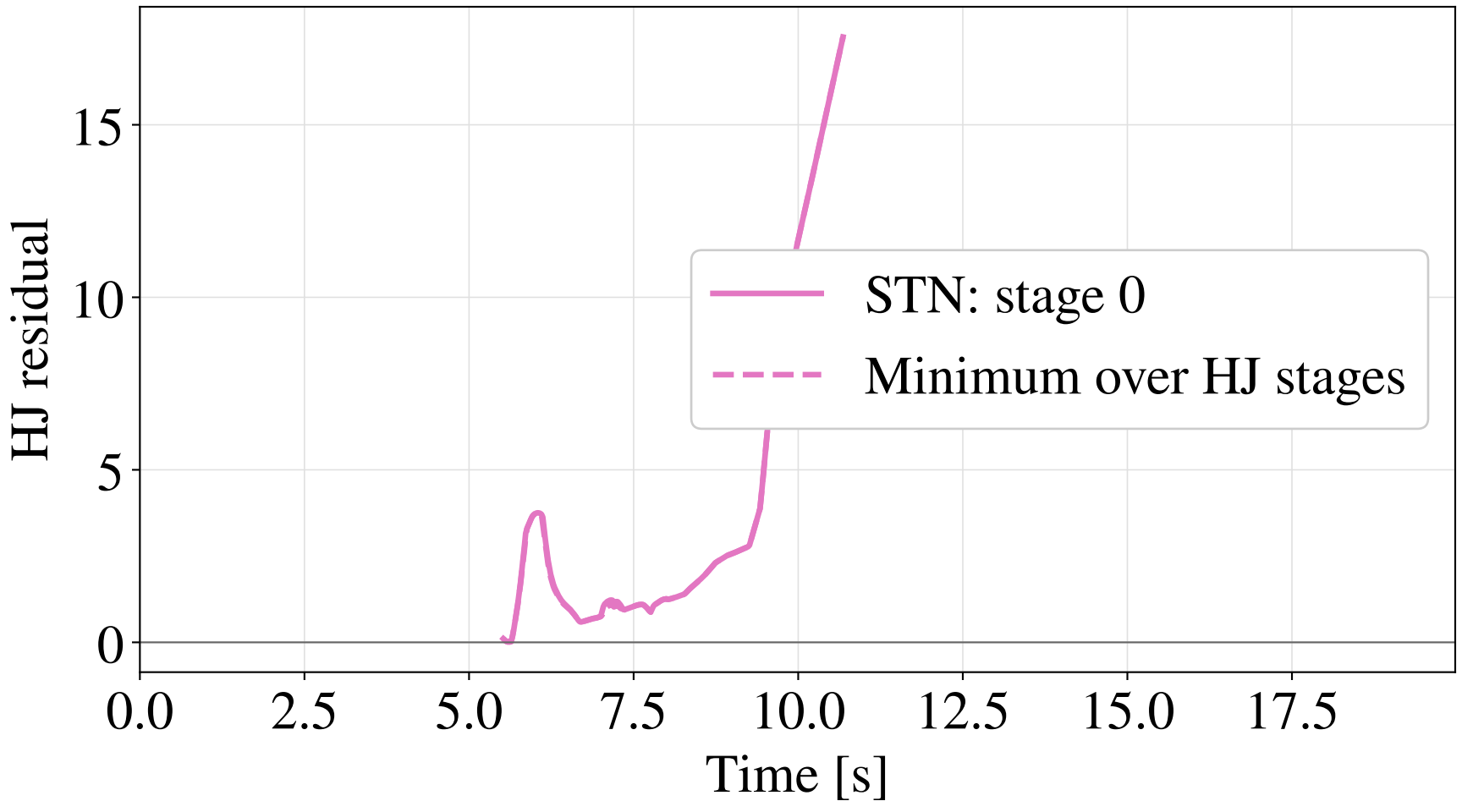

1e−8
2
1
0
HJ slack
Euclidean: stage 0
Maximum over HJ stages
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]

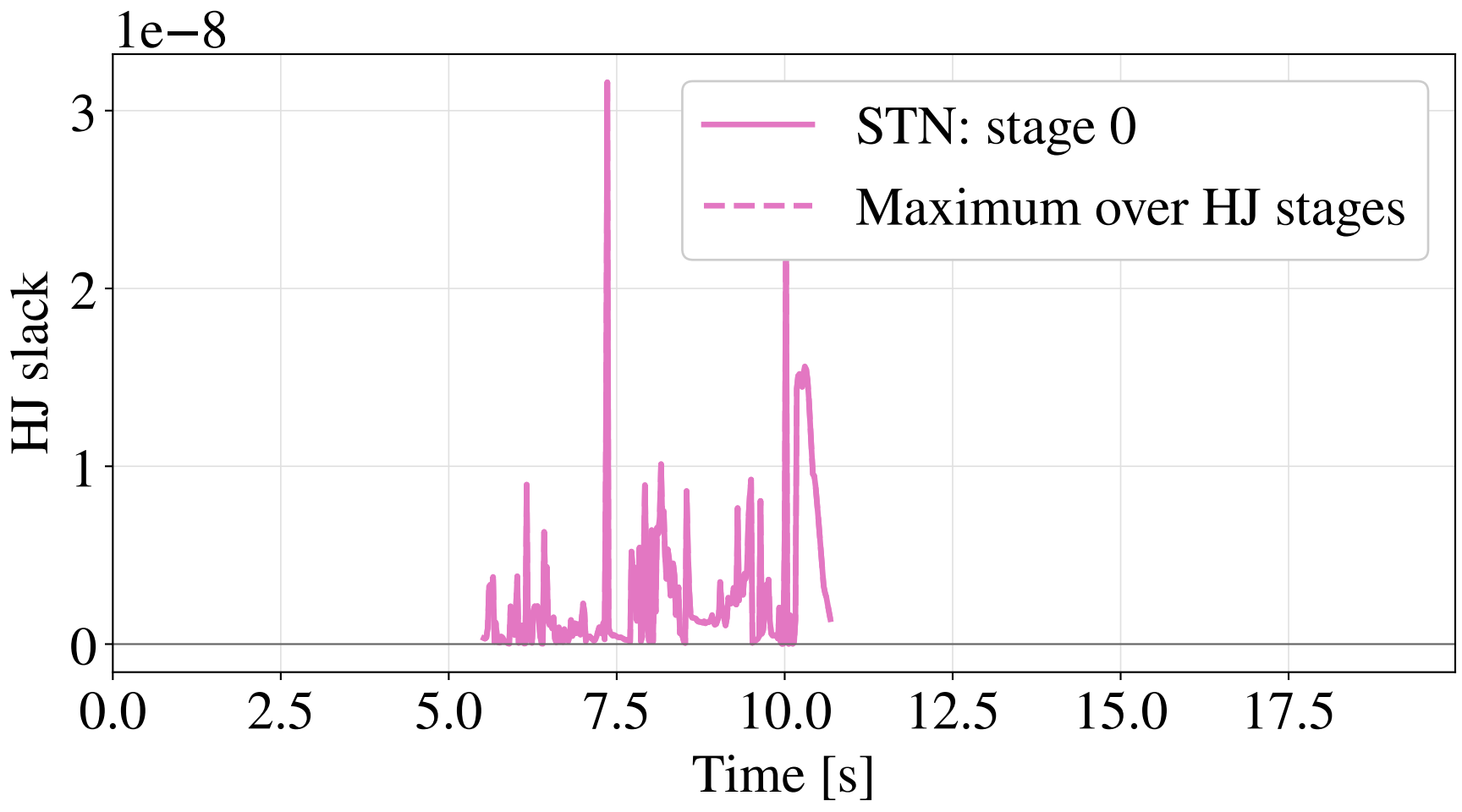

HJ residual
15
10
5
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]
DCE: stage 0
Minimum over HJ stages

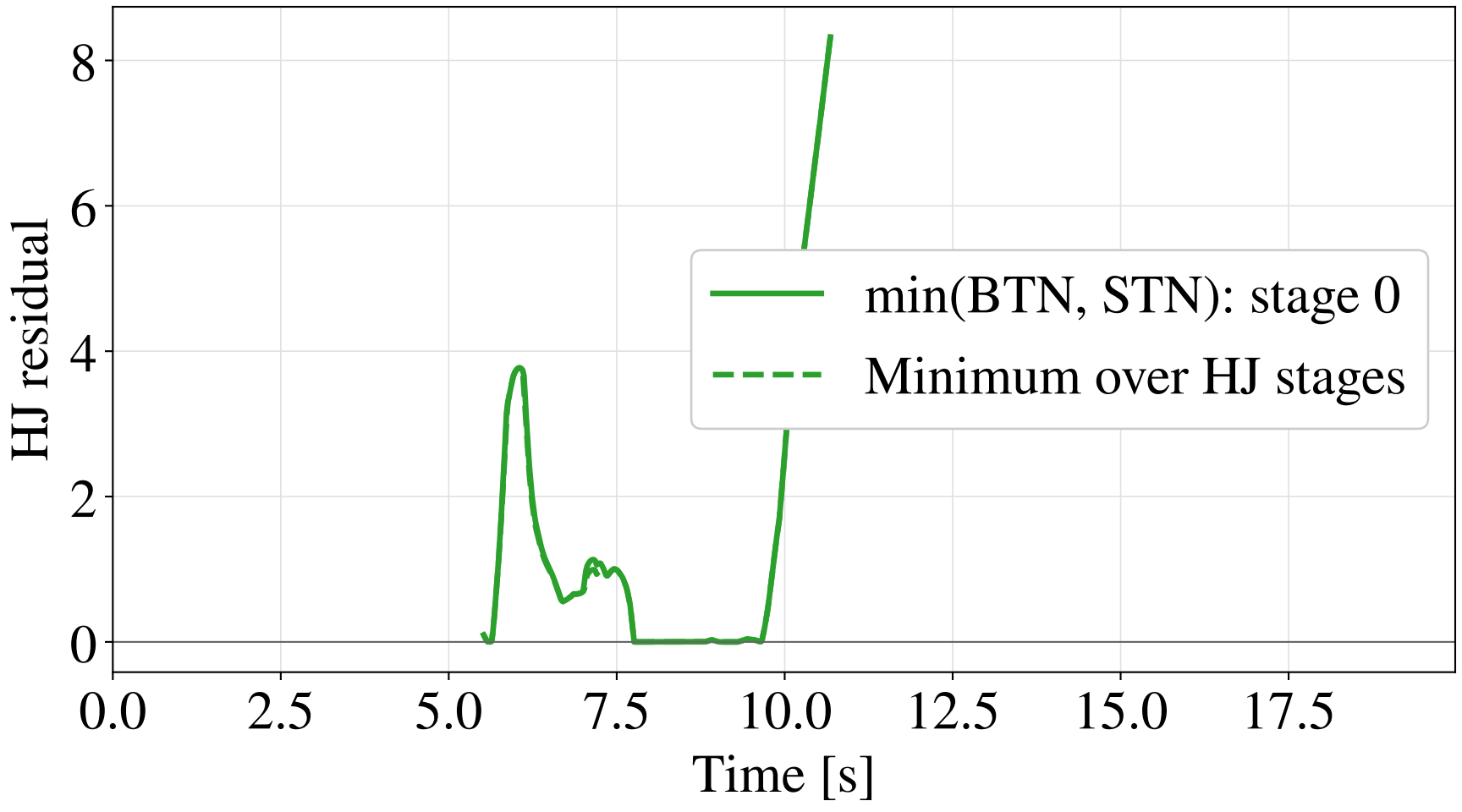

1e−8
3
2
1
0
HJ slack
DCE: stage 0
Maximum over HJ stages
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]

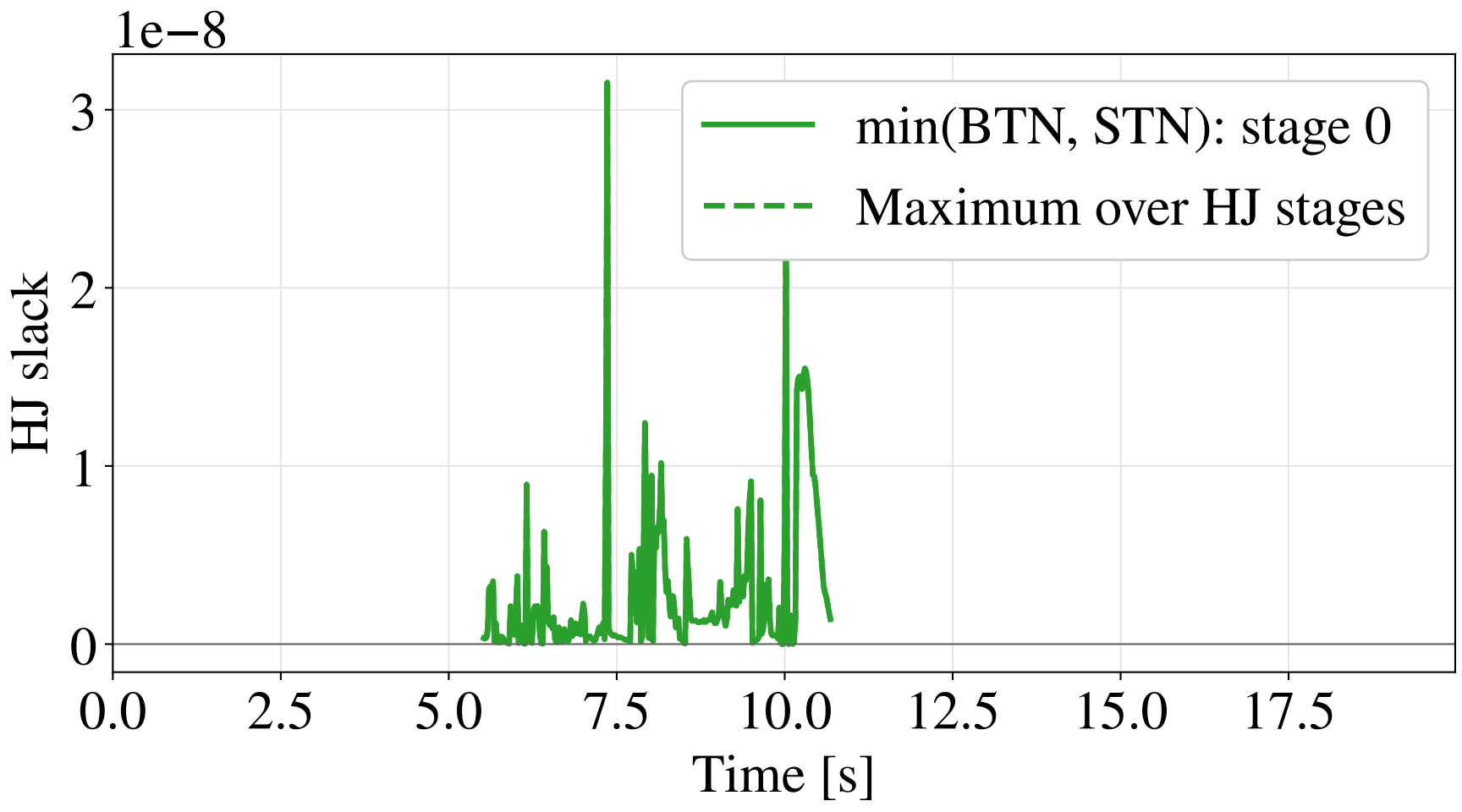

HJ residual
8
6
4
2
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]
BTN: stage 0
Minimum over HJ stages

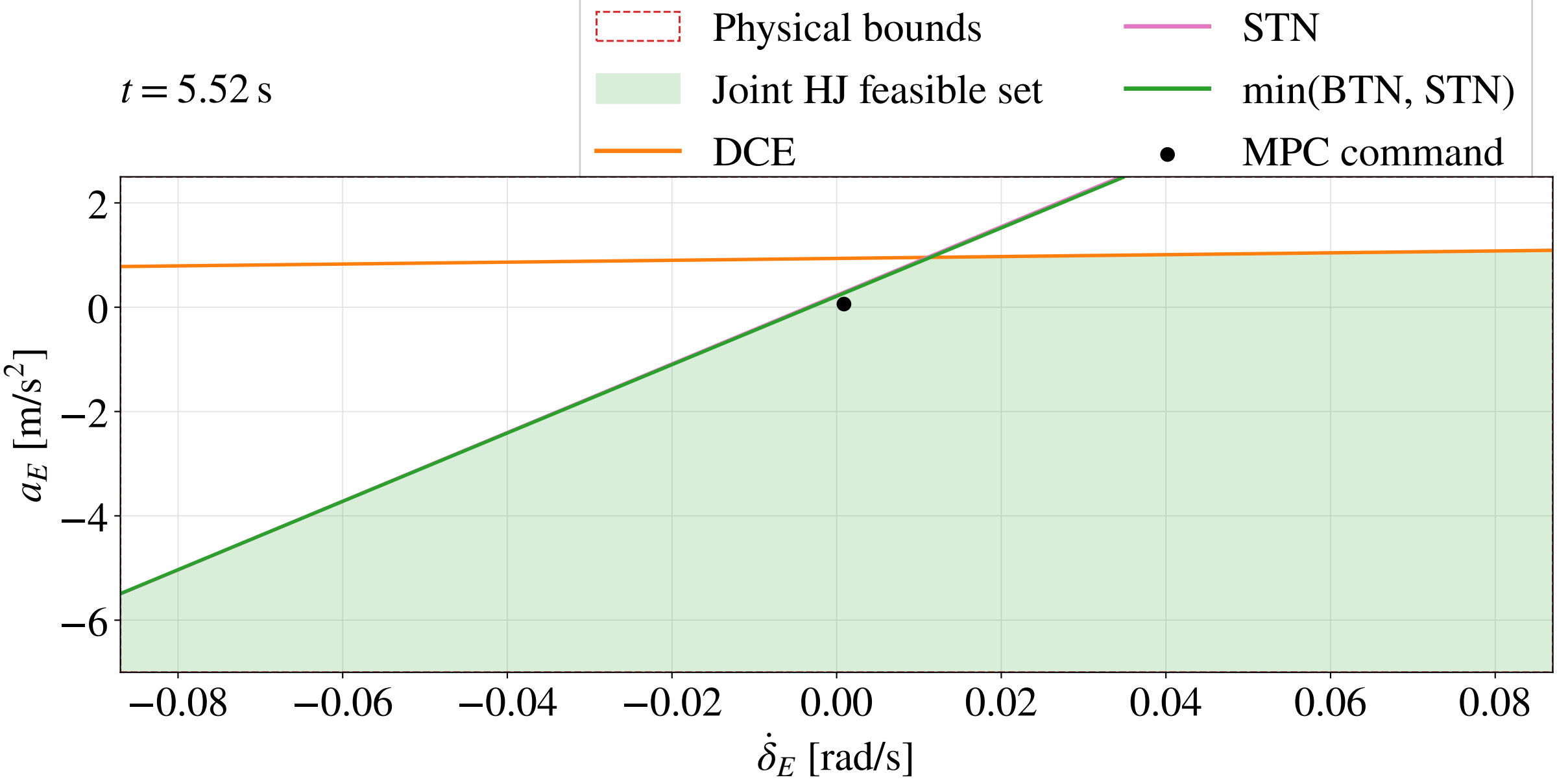

1e−8
3
2
1
0
HJ slack
BTN: stage 0
Maximum over HJ stages
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]

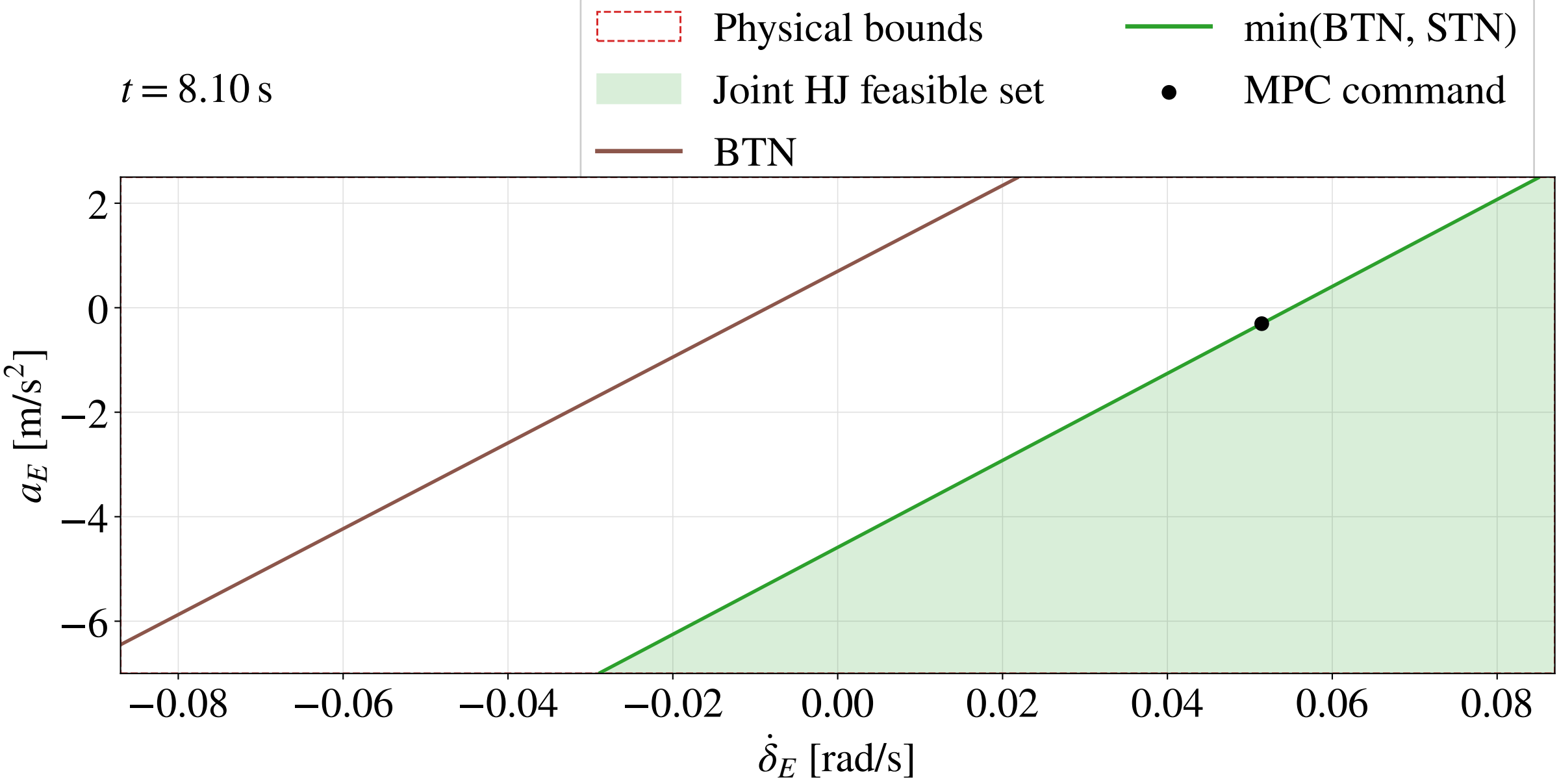

HJ residual
15
10
5
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]
STN: stage 0
Minimum over HJ stages

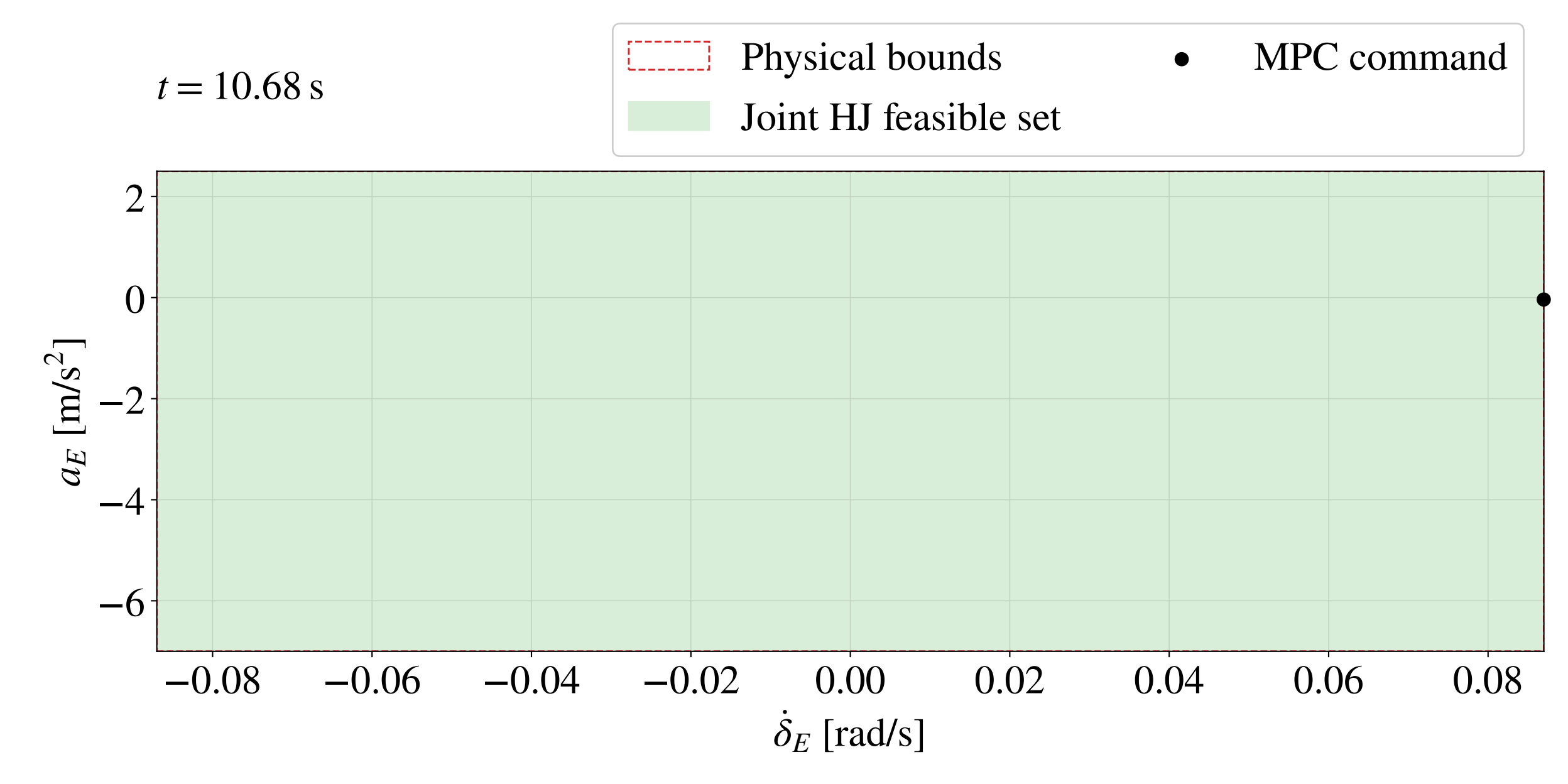

1e−8
3
2
1
0
HJ slack
STN: stage 0
Maximum over HJ stages
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]


HJ residual
8
6
4
2
0
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]
min(BTN, STN): stage 0
Minimum over HJ stages


1e−8
3
2
1
0
HJ slack
min(BTN, STN): stage 0
Maximum over HJ stages
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
Time [s]


$t = 5.52$ s
Physical bounds
Joint HJ feasible set
DCE
STN
min(BTN, STN)
MPC command
2
0
−2
−4
−6
a_E [m/s^2]
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
$\dot{\delta}_E$ [rad/s]


Physical bounds
Joint HJ feasible set
BTN
min(BTN, STN)
MPC command
$t = 8.10$ s
2
0
−2
−4
−6
a_E [m/s^2]
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
$\dot{\delta}_E$ [rad/s]


$t = 10.68$ s
Physical bounds
Joint HJ feasible set
MPC command
2
0
−2
−4
−6
a_E [m/s^2]
−0.08
−0.06
−0.04
−0.02
0.00
0.02
0.04
0.06
0.08
$\dot{\delta}_E$ [rad/s]